PHYSICS

1) Time period of oscillation (t) of a wire under tension depends on length of wire (□), mass per unit length (m) and tension (T) then correct option for time period of wire is -

(1) $□^{1/2}\ T^{1/2}\ m^{-1/2}$
(2) $□^{1/2}\ T^{-1/2}\ m^{1/2}$
(3) $□^{1}\ T^{-1/2}\ m^{1/2}$
(4) $□^{0}\ T^{-1/2}\ m^{1/2}$

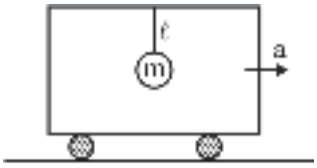
2) A projectile has same range R for two angles of projection. If t_1 & t_2 be the time of flight for the two cases then :

(1) $R=\frac{gt_1t_2}{2}$
(2) $R=\frac{g(t_1+t_2)^2}{2}$
(3) $R=g\sqrt{t_1t_2}$
(4) $R=2g\frac{t_1t_2}{t_1+t_2}$

3) Two particle A and B are located in x-y plane at points (0, 0) m and (0, 4) m. They simultaneously start moving with velocities $\vec{v}_A = 2\hat{j}$ m/s and $\vec{v}_B = 2\hat{i}$ m/s . Select the correct alternative(s) :
(A) The distance between them is constant
(B) The distance between them first decreases and then increases
(C) The shortest distance between them is $2\sqrt{2}$ m
(D) Time after which they are at minimum distance is t = 1s

(1) Only B is correct
(2) Only C is correct
(3) A, B,C are correct
(4) B,C,D are correct

4) A pendulum of mass m & length □ is suspended from the ceiling of a trolley which has a constant acceleration 'a' in the horizontal direction as shown in figure. Work done by the tension is (with respect to trolley) :-



(1) –mg□(1 – cosθ)
(2) ma□sinθ

(3) ma□cosθ

(4) zero

5)

During regulation action of a Zener diode, correct statement is :-
(a) the current and voltage across the Zener remains fixed.
(b) the current through the series resistance (R_s) changes.
(c) the Zener resistance is constant.
(d) the resistance offered by the Zener diode changes.

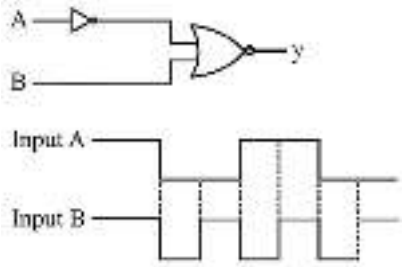
(1) a and c

(2) a and d

(3) b and c

(4) b and d

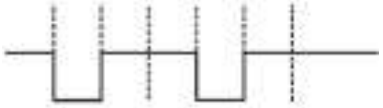
6) The logic circuit shown below has the input waveforms 'A' and 'B'. Pick out the correct output waveform :-



(1)



(2)



(3)



(4)



7) If an interference pattern have maximum and minimum intensities in 36 : 1 ratio, then what will be the ratio of amplitudes of interfering waves ?

(1) 2 : 7

(2) 7 : 3

(3) 4 : 7

(4) 7 : 5

8) Select correct option regarding photo electric effect :

(1) Photoelectric current is directly proportional to the frequency of light.

(2) Stopping potential is related to average frequency in case of polychromatic light is used

(3) Stopping potential is discriminator between wave nature and quantum nature of light.

(4) Kinetic energy of fastest electron is directly proportional to the frequency of light.

9) In a fission reaction

$$^{239}_{94}Pu \rightarrow {}^{139}X + {}^{100}Y + Q$$

The binding energy per nucleon of X and Y is 8.4 MeV whereas of Pu^{239} is 7.5 MeV. The total energy released will be :-

(1) 215.1 MeV

(2) 352.2 MeV

(3) 415.2 MeV

(4) 125.3 MeV

10) Li nucleus has three protons and four neutrons. Mass of lithium nucleus is 7.016005 u. Mass of proton is 1.007277 u and mass of neutron is 1.008665 u. Mass defect for lithium nucleus (in u) is :-

(1) 0.04048

(2) 0.04062

(3) 0.04052

(4) 0.04055

11) The ionisation energy of Li^{++} atom in ground state is equal to :-

(1) 13.6 × 9eV

(2) 13.6 J

(3) 13.6 erg

(4) 13.6×10^{-19} J

12) The focal lengths of the objective and the eye-piece of a compound microscope are 1 cm and 5 cm respectively. An object is placed at a distance of 1.1 cm from the objective. If the final image is formed at the least distance of distinct vision, the magnifying power is :-

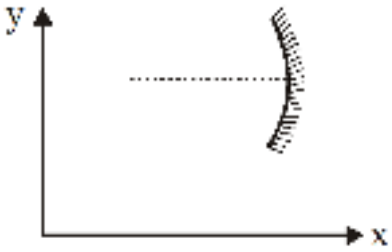
(1) 60

(2) 50

(3) 40

(4) none of these

13) Figure shows a concave mirror with its principal axis parallel to x–axis and focus on the y–axis. The centre of curvature is at (–108 cm, 54 cm). The coordinates of the image of a point source A (81



cm, 48 cm) will be :-

(1) (144 cm, 46 cm)

(2) (144 cm, 62 cm)

(3) (36 cm, 8 cm)

(4) None of these

14) A thin glass prism of refractive index 1.5 produces a deviation of 4° of a ray incident at a small angle.What will be the deviation of the same incident ray by the same prism if it is immersed in water of refractive index 4/3?

(1) 1°

(2) 2°

(3) 8°

(4) 16°

15) An electron having kinetic energy K is moving in a circular path of radius R perpendicular to a uniform magnetic field. If the kinetic energy is doubled and magnetic field tripled, the radius of path will become :-

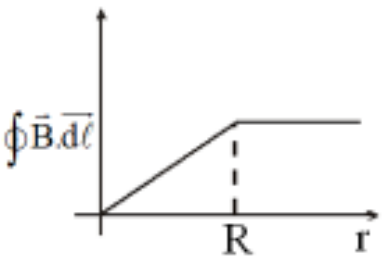
(1) $\frac{2R}{3}$

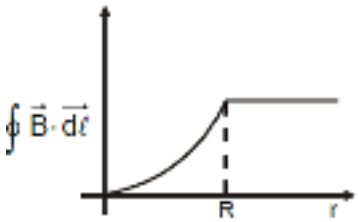
(2) $\frac{\sqrt{2}}{3}R$

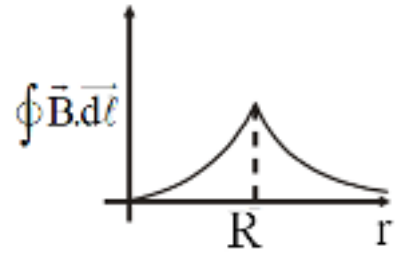
(3) $\sqrt{\frac{2}{3}}R$

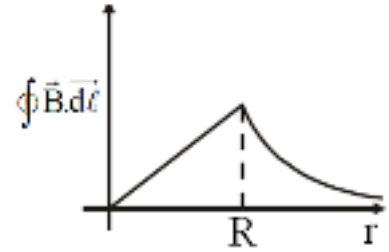
(4) $\frac{2}{\sqrt{3}}R$

16) A cylindrical wire of radius R is carrying current i uniformly distributed over its cross-section. If a circular loop of radius 'r' is taken as amperian loop, then the variation value of $\oint \vec{B}.d\vec{\ell}$ over this loop with radius ' r ' of loop will be best represented by:

(1) 

(2) 

(3) 

(4) 

17) A 50 W, 100 V lamp is connected in series with a capacitor of capacitance $\frac{50}{\pi\sqrt{x}}\mu F$ and this combination is connected with 200 V, 50Hz AC source. The value of x will be ____ .

(1) 5

(2) 3

(3) $\frac{1}{3}$

(4) 4

18) **Assertion :** In EMW spectrum there is no sharp division between one kind of the wave and the next.
Reason : The classification of EMW spectrum is based roughly on how the waves are produced and detected.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Assertion is false but Reason is true.

19) About 5% of the power of a 200 W is converted into visible radiation, then intensity of visible radiation at distance 10 m is.

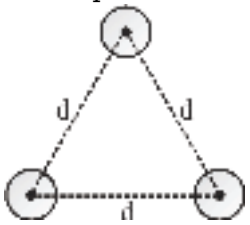
(1) 0.4 W/m^2

(2) 0.004 W/m^2

(3) 0.08 W/m^2

(4) 0.008 W/m^2

20) Three solid spheres, each of mass m and radius R, are released from the position shown in figure. The speed of any one sphere at the time of collision would be :



(1) $\sqrt{Gm\left(\frac{1}{d}-\frac{3}{R}\right)}$

(2) $\sqrt{Gm\left(\frac{3}{d}-\frac{1}{R}\right)}$

(3) $\sqrt{Gm\left(\frac{2}{R}-\frac{1}{d}\right)}$

(4) $\sqrt{Gm\left(\frac{1}{R}-\frac{2}{d}\right)}$

21) Body A of mass 4m moving with speed u collides with another body B of mass m, at rest. The collision is head on and elastic in nature. After the collision, the fraction of energy lost by the colliding body A is :

(1) $\frac{8}{9}$

(2) $\frac{16}{25}$

(3) $\frac{4}{25}$

(4) $\frac{5}{9}$

22) If T is the surface tension of a liquid, the energy needed to break a liquid drop of radius R into 64 drops is :-

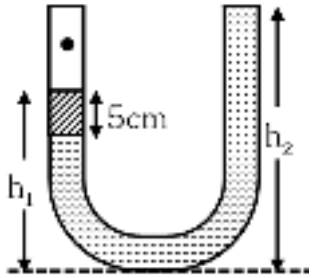
(1) $6\pi R^2T$

(2) πR^2T

(3) $12\pi R^2T$

(4) $8\pi R^2T$

23) An open ended U–tube of uniform cross–sectional area contains water (ρ = 1.0 gm/cm^3) standing initially 30 cm from the bottom in each arm. An immiscible liquid of density 4.0 gm/cm^3 is added to one arm until a layer of 5 cm high forms as shown in figure below. Find ratio $\frac{h_2}{h_1}$:-



(1) $\frac{3}{1}$

(2) $\frac{5}{2}$

(3) $\frac{2}{1}$

(4) $\frac{8}{5}$

24) A horizontal disc rotates freely about a vertical axis through its centre. A ring, having the same mass and radius as the disc, is now gently placed on the disc. After some time, the two rotate with a common angular velocity-

(a) Some friction exists between the disc and the ring.

(b) The angular momentum of the 'disc plus ring' is conserved.

(c) The final common angular velocity is $\frac{2}{3}$rd of the initial angular velocity of the disc.

(d) $\frac{2}{3}$rd of the initial kinetic energy changes to heat.

Which of the above statements are true?

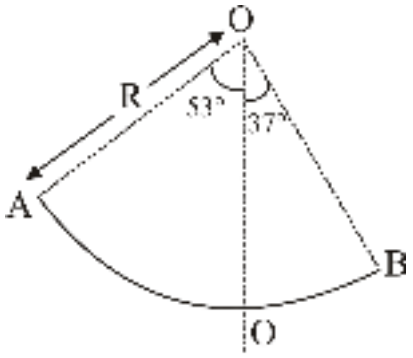
(1) a, b, d

(2) b, c, d

(3) c, d, a

(4) a, c

25) A section of fixed smooth circular track of radius R in vertical plane is shown in the figure. A block is released from position A and leaves the track at B. The radius of curvature of its trajectory,



when it just leaves the track at B, is-

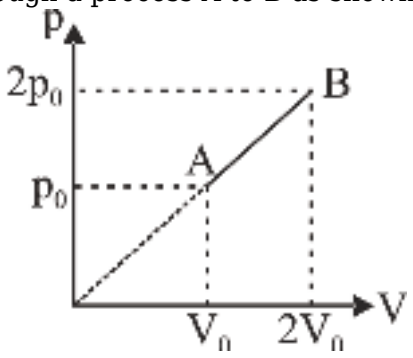
(1) R

(2) R/4

(3) R/2

(4) R/8

26) An enclosed one mole of an monoatomic gas is taken through a process A to B as shown in figure. The molar heat capacity of the gas for this process is



(1) R

(2) 2R

(3) 3R

(4) 4R

27) If $P = av^2$, then find work done in changing volume v_1 to v_2.

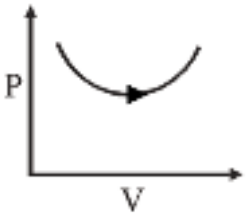
(1) $\frac{a\left(v_2^2 - v_1^2\right)}{2}$

(2) $\frac{a\left(v_2^3 - v_1^3\right)}{3}$

(3) $a(v_2 - v_1)$

(4) $\frac{a\left(v_2^4 - v_1^4\right)}{4}$

28) Consider the process on a system shown in the figure. During the process, work done by the system :-



(1) Continuously increases

(2) Continuously decreases

(3) First increases then decreases

(4) First decreases then increases

29) **Assertion :** When a girl sitting on a swing stand up, the periodic time of the swing will decrease.
Reason : In standing position of a girl, the effective length of the swing will decrease.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

30) A particle is moving along the x-axis under the influence of a force given by F = –5x + 15. At t = 0, the particle is located at x = 6 and is having zero velocity. It takes 0.5 seconds to reach the origin for the first time. The equation of motion of the particle can be represented by

(1) $x = 3 + 3 \cos \pi t$

(2) $x = 3 \cos \pi t$

(3) $x = 3 + 3 \sin \pi t$

(4) $x = 3 + 3 \cos (2 \pi t)$

31) Two wires are in unison. If the tension in one of the wires is increased by 2%, 5 beats are produced per second. The initial frequency of each wire is :

(1) 200 Hz

(2) 400 Hz

(3) 500 Hz

(4) 1000 Hz

32)

An electric dipole of length 1 cm is placed with its axis making an angle of 30° to an electric field of strength 10^4 NC^{-1}. If it experiences a torque of $10\sqrt{2}$ Nm, the potential energy of the dipole is :

(1) 0.245 J

(2) 2.45×10^{-4} J

(3) 0.0245 J

(4) 24.5 J

33) Two metals plates are placed in parallel. If capacitance of capacitor is C, then what will be potential energy stored in the region between two plates :-



(1) $\frac{1}{2}\frac{Q^2}{C}$

(2) $\frac{9}{2}\frac{Q^2}{C}$

(3) $\frac{3}{2}\frac{Q^2}{C}$

(4) zero

34) The efficiency of a cell, when connected to a resistance R, is 60%. What will be its efficiency, if the external resistance is increased to six times?

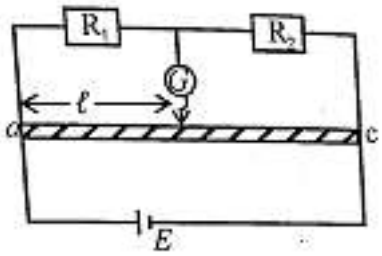
(1) 80 %

(2) 90%

(3) 55%

(4) 95%

35) The given figure shows rough sketch of meter bridge. (G) deflects zero at length cm. Now, R_1 and R_2 are interchanged, then balancing length increases by 25 cm. Find R_1/R_2



(1) $\frac{3}{5}$

(2) $\frac{2}{5}$

(3) $\frac{2}{3}$

(4) $\frac{5}{2}$

36) Find the current in a copper wire having cross sectional area 2 mm^2.
(Given $E = 8.5 \times 10^{-3}$ V/m,
$\rho = 1.7 \times 10^{-8}$ Ω-m)

(1) 1 A

(2) 3 A

(3) 2 A

(4) 4 A

37) If displacement of a body S = (200 ± 5) metres and time taken by it is t = (20 ± 0.2) seconds, then percentage error in velocity -

(1) 2.5%

(2) 3.5%

(3) 4.5%

(4) 5.5%

38) In a vernier callipers, 10 divisions of vernier scale coincides with 9 divisions of main scale. one division of main scale is of 0.1 cm. If in the measurement of inner diameter of cylinder zero of vernier scale lies between 1.3 cm and 1.4 cm of main scale and 2nd division of vernier scale coincides with main scale division then diameter will be:-

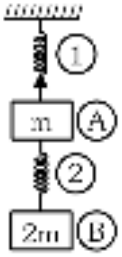
(1) 1.30 cm

(2) 1.34 cm

(3) 1.32 cm

(4) 1.36 cm

39) Initially system was in equilibrium, just after cutting the spring (1) acceleration of (A) is a and acceleration of (B) is b then value of a and b :



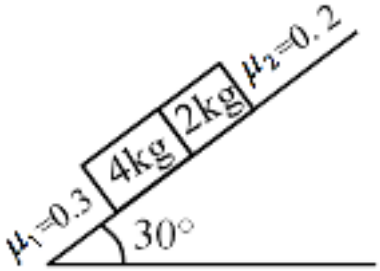
(1) g↓, g↓

(2) g↓, g↑

(3) 3g↓, 0

(4) g↑, g↓

40) Blocks are sliding down on a inclined plane, acceleration of 2 kg will be :



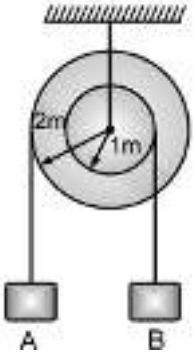
(1) 1.6 m/s^2

(2) 2.6 m/s^2

(3) 3.6 m/s^2

(4) 4.6 m/s^2

41) In the pulley system (stick to each other) shown, if radii of the bigger and smaller pulley are 2 m and 1 m respectively and the acceleration of block A is 5 m/s^2 in the downward direction, then the acceleration of block B will be:



(1) 0 m/s^2

(2) 5 m/s^2

(3) 10 m/s^2

(4) 5/2 m/s^2

42) Two bodies of different materials and same surface area, coated by similar material, are heated

to same temperature. They are allowed to cool down in same surroundings. If $\frac{dQ}{dt}$ represents rate of loss of heat and $\frac{d\theta}{dt}$ represents rate of fall of temperature, then for both

(1) $\frac{dQ}{dt}$ is same but $\frac{d\theta}{dt}$ is different

(2) $\frac{dQ}{dt}$ is different but $\frac{d\theta}{dt}$ is same

(3) $\frac{d\theta}{dt}$ as well as $\frac{dQ}{dt}$ is same

(4) $\frac{d\theta}{dt}$ as well as $\frac{dQ}{dt}$ is different.

43) An air capacitor C connected to a battery of e.m.f. V acquires a charge q and energy E. The capacitor C is disconnected from the battery and a dielectric slab is placed between the plates. Which of the following statements is correct?

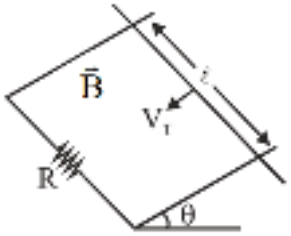
(1) V and q decreases but C and E increases

(2) V remains unchanged, but q, E and C increases

(3) q remains unchanged, C increases, V and E decreases

(4) q and C increases but V and E decreases

44) A copper rod of mass m slides under gravity on two smooth parallel rails distance apart set at an angle θ to the horizontal. At the bottom, the rails are joined by a resistance R.



There is a uniform magnetic field perpendicular to the plane of the rails. The terminal velocity of the rod is

(1) $\frac{mgR\cos\theta}{B^2\ell^2}$

(2) $\frac{mgR\sin\theta}{B^2\ell^2}$

(3) $\frac{mgR\tan\theta}{B^2\ell^2}$

(4) $\frac{mgR\cot\theta}{B^2\ell^2}$

45) Young's Double Slit Experiment is first performed in air then in a medium other than air. It is found that 10th bright fringe in medium lies where 8th dark fringe lies in air. The refractive index of medium is nearly :-

(1) 1.59

(2) 1.78

(3) 1.69

(4) 1.33

CHEMISTRY

1) For true statement use 'T' for false use 'F' :-
(a) XeF_4 polar as well as planar.
(b) Bond length order ⇒ Si – O > Cl – O
(c) In urea inter molecular H-bonding is not possible
(d) Covalent character :- $SiCl_4$ > CuCl > NaCl
(e) Melting point ⇒ HI > HBr > HCl

(1) F, T, F, F, F

(2) T, F, F, T, T

(3) F, T, F, T, T

(4) T, F, T, F, T

2) Which of the following molecule has single super pair:-

(1) H_2O_2

(2) C_2H_6

(3) C_2H_4

(4) N_2H_4

3)

Considering ethene molecule, which of the following statements is/are correct ?
I. C–C bond is formed by overlapping of sp^2–sp^2 hybrid orbitals.
II. C–H bond is formed by overlapping of sp^2–s orbitals.
III. pπ–pπ bond is present.
IV. H–C–H bond angle is greater than H–C–C bond angle.

(1) I, II, III

(2) I, II, III, IV

(3) I, II

(4) I, II, III

4)

Which of the following statement is correct :-

(1) $[Pt(NO_3)_2(en)_2]^{2+}$ complex ion can show linkage isomerism

(2) $[Cr(CO_3)(NH_3)_4]Br$ produces white ppt. on reaction with $AgNO_3$

(3) $FeCl_2.6H_2O$ can show hydrate isomerism

(4) $[FeCl_3(NH_3)_3]$ can exhibit structural isomerism

5) Arrange the following in order of decreasing number of unpaired electrons :-
I. $[Fe(H_2O)_6]^{2+}$ II. $[Cr(NH_3)_6]^{+3}$
III. $[Ni(NH_3)_6]^{+2}$ IV. $[FeF_6]^{-3}$

(1) IV, I, II, III

(2) I, II, III, IV

(3) III, II, I, IV

(4) II, III, I, IV

6) Match the column correctly ?

	Column-I (Coordination entity)		**Column-II (Color)**
(a)	$[Cu(NH_3)_4]^{+2}$	(P)	Purple
(b)	$[Co(NH_3)_6]^{+3}$	(Q)	Blue
(c)	$[Ni(H_2O)_6]^{+2}$	(R)	Yellow
(d)	$Na_4[Fe(CN)_5NOS]$	(S)	Green

(1) a→R, b→S, c→P, d→Q

(2) a→S, b→Q, c→R, d→P

(3) a→Q, b→R, c→S, d→P

(4) a→Q, b→R, c→P, d→S

7) Correct order of oxidising nature is :-

(1) $MnO_4^- < TcO_4^- < ReO_4^-$

(2) $CrO_4^{-2} > MoO_4^{-2} > WO_4^{-2}$

(3) $F_2 < Cl_2 < Br_2 < I_2$

(4) $PbO > PbO_2$

8) Compounds A and B are treated with dilute HCl seperately. The gases liberated are Y and Z respectively Y turns acidified dichromate paper green while Z turns lead acetate paper black. So A and B compounds are respectively :-

(1) Na_2SO_3, Na_2S

(2) NaCl, Na_2CO_3

(3) Na_2S, Na_2SO_3

(4) Na_2SO_3, K_2SO_4

9) Which one is incorrect for actinoids ?

(1) All are radioactive in nature

(2) Highest oxidation state is +8

(3) In boiling water form mixture of oxide and hydride

(4) 5f orbital are less buried than 4f

10) On moving from Ce^{+3} to Lu^{+3} the cation having maximum no. of unpaired electrons is :-

(1) Ce^{+3}

(2) Lu^{3+}

(3) Eu^{+3}

(4) Gd^{+3}

11) Identify the **incorrect** statement :

(1) Boron trioxide is acidic, aluminium and gallium trioxides are amphoteric while indium and thallium trioxides are basic.

(2) The hybridisation of Al in $[Al(H_2O)_6]^{3+}$ is d^2sp^3.

(3) Two isotopes of Boron, namely $_{10}B$(19%) and $_{11}B$(81%) are known.

(4) The relative stability of group 13 elements in +1 oxidation state varies as Al < Ga < In < Tl.

12) Choose correct statement.
(A) Polonium occurs in nature as a decay product of thorium and uranium minerals.
(B) Oxygen is the most abundant of all the elements on earth.
(C) Polonium rarely show '–2' oxidation state.

(1) Only A and C

(2) Only B and C

(3) Only C

(4) A, B and C

13) Which of the following statement is incorrect :-

(1) Amongst Li^- , Be^-, B^- and C^- , Be^- is the least stable ion

(2) $I.E_1$ of Sr > Rb but $I.E_{II}$ for Rb > Sr

(3) Lithium is an alkali metal

(4) $_{27}M^{+n}$ ion (Magnetic moment = 4.89 BM) has +2 oxidation state

14) Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : Ionisation enthalpy increases along each series of the transition elements from left to right. However, small variations occur.
Reason (R) : There is corresponding increase in nuclear charge which accompanies the filling of electrons in the inner d-orbitals.
In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) **(A)** is correct but **(R)** is not correct.

(2) **(A)** is not correct but **(R)** is correct.

(3) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**.

(4) Both **(A)** and **(R)** are correct but **(R)** is **not** the correct explanation of **(A)**.

15) In salt analysis, Cl^-, Br^- and I^- are identified by adding dilute HNO_3 followed by $AgNO_3$ solution to the salt solution. Choose the **correct** statements from the following :

A. Cl^- gives pale yellow precipitate.

B. Br^- gives pale yellow precipitate which is partially soluble in NH_4OH.

C. I^- gives yellow precipitate which is insoluble in NH_4OH.

D. Cl^- gives white precipitate which is insoluble in NH_4OH.

E. I^- gives purple violet gas.

Choose the **correct** answer from the options given below :

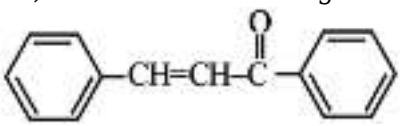
(1) A and B only

(2) A and E only

(3) D and E only

(4) B and C only

16) IUPAC name of the given compound



(1) 1, 3-diphenylprop-2-en-1-one

(2) 1, 3-diphenylprop-1-en-3-one

(3) 1, 3-diphenylpropanone

(4) None of these

17) Find the relation between compounds :-



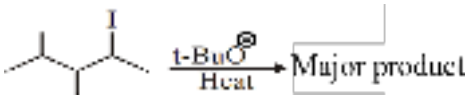
(1) Enantiomers

(2) Homomers

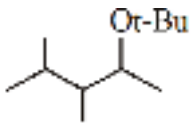
(3) Meso compound

(4) Diastereomers

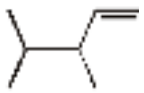
18) The major product in the following reaction is :

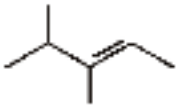


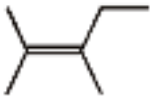
(1)



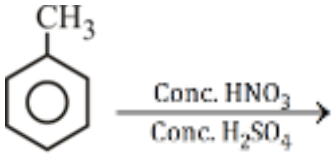
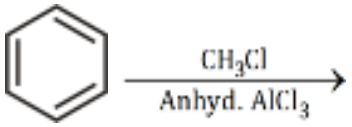
(2)



(3) 

(4) 

19) Match the following :-

(a)		(p)	Kolbe's electrolysis
(b)	CH_3–COOK $\xrightarrow{\text{electrolysis}}$	(q)	Nitration
(c)	CH_3–Br $\xrightarrow[\text{Dryether}]{\text{Na}}$	(r)	Wurtz reaction
(d)		(s)	Friedal-craft alkylation

(1) a–p, b–q, c–r, d–s

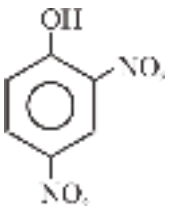
(2) a–q, b–r, c–p, d–s

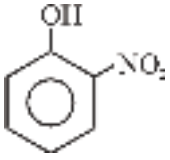
(3) a–q, b–p, c–s, d–r

(4) a–q, b–p, c–r, d–s

20) Which of the following compound can react with Na, NaOH and $NaHCO_3$:-

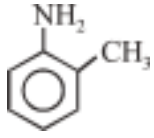
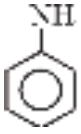
(1) 

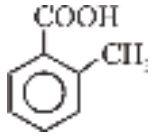
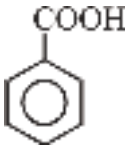
(2) 

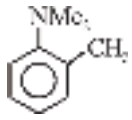
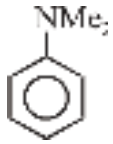
(3) 

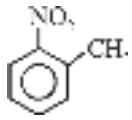
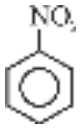
(4) All of these

21) Select the correct statement :

(1)  is more basic than 

(2)  is less acidic than 

(3)  is more basic than 

(4)  has shorter C–N bond length than that of 

22) In carius method of sulphur estimation 0.192 g of an organic compound gave 0.466 g of barium sulphate what is the percentage of sulphur in the compounds :-

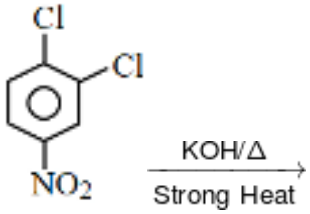
(1) 42 %

(2) 50 %

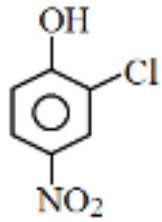
(3) 25 %

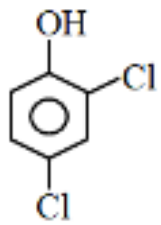
(4) 33 %

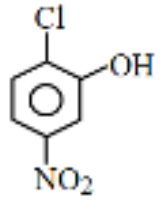
23) Product of the following given reaction is :-

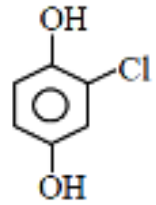


Major product

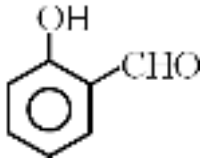
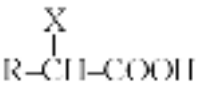
(1) 

(2) 

(3) 

(4) 

24) Match the List-I with List-II :

List-I (Reaction)		List-II (Product)	
(a)	Rosenmund reaction	(i)	
(b)	HVZ reaction	(ii)	R–I
(c)	Reimer-Tiemann reaction	(iii)	
(d)	Finkelstein reaction	(iv)	R–CHO

Select the correct option :

(1) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)

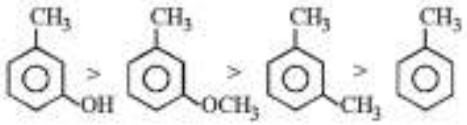
(2) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

(3) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)

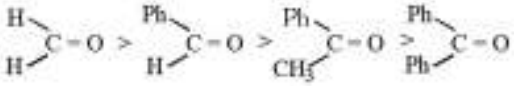
(4) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

25)

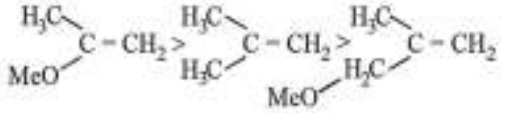
Which of the following order is correct :-

(a) 

(Rate of ESR)

(b) 

(Rate of NAR)

(c) 

(Rate of EAR)

(d) $CH_2=CH-CH_3 > CH_2=CH_2 > CH_2=CH-Cl$

(Rate of EAR)

(1) Only a, b, c

(2) Only b, c

(3) Only b, c, d

(4) a, b, c, d

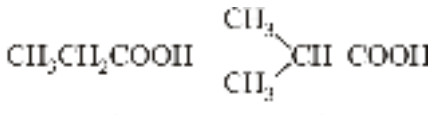
26) Identify correct statement -

(1) Aldehyde and ketone having at least one α-hydrogen undergo a reaction in the presence of dilute alkali as a catalyst to form β-hydroxy aldehyde or ketone (Aldol or ketol) called aldol condensation.

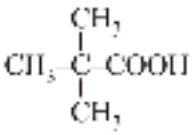
(2) Aldehyde which do not have an α-hydrogen undergo disproportionation reaction on heating with conc. alkali is called cannizzaro reaction.

(3) Carboxylic acid having α-hydrogen are halogenated at the α-position on treatment with chlorine or bromine in presence of small amount of red P to give α-halo carboxylic acid is called swart's reaction.

(4) Nitriles are reduced to corresponding imine with Hydrogen gas in presence of paladium, which on hydrolysis give corresponding aldehyde is called stephen reaction.



(I) (II)



(III)

27) Among these acids which of the following order is correct for ease of esterification with MeOH:-

(1) I > II > III

(2) III > II > I

(3) II > I > III

(4) I > III > II

28) ***Statement -1 :-*** During the S_N1 reaction of alkyl halide, step-I is slowest and reversible.
Statement-2 :- Step-I involves the C–X bond breaking for which the energy is obtained through solvation of halide ion with the proton of protic solvent.

(1) Both statements are correct.

(2) Statement-1 is correct statement-2 is incorrect.

(3) Statement-1 is incorrect statement-2 is correct.

(4) Both statements are incorrect.

29) Given below are two statements :
Statement I : Primary aliphatic amines react with HNO_2 to give unstable diazonium salts.
Statement II : Primary aromatic amines react with HNO_2 to form diazonium salts which are stable even above 300 K.
In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both **Statement-I** and **Statement-II** are incorrect.

(2) **Statement-I** is correct but **Statement-II** is incorrect.

(3) **Statement-I** is incorrect but **Statement-II** is correct.

(4) Both **Statement-I** and **Statement-II** are correct.

30) Which of the following statements is not true about glucose.

(1) It is an aldohexose

(2) On heating with HI it forms n-hexane

(3) It is present in furanose form

(4) It does not give 2,4-DNP test

31) The pH and conductance of 1M weak acid (HX) was found to be 5 and 4×10^{-5} S, respectively. The conductance was measured using a cell where the electrode plates have a surface area of 1 cm^2 and at a distance of 15 cm apart. The value of the limiting molar conductivity is ..... S m^2 mol^{-1}. (nearest integer)
(Given : degree of dissociation of the weak acid (α) << 1)

(1) 6×10^{-3}

(2) 6

(3) 5×10^{-3}

(4) 5

32) Given,
(A) n = 5, $m_{\square}$ = –1

(B) n = 3, $\square$ = 2, $m_{\square}$ = –1, $m_s = +\frac{1}{2}$
The maximum number of electron(s) in an atom that can have the quantum numbers as given in (A) and (B) respectively are :

(1) 26 and 1

(2) 4 and 1

(3) 2 and 4

(4) 8 and 1

33) When the combustion of one mole of heptane (□) is carried out at a temperature T (298 K), The difference between ΔH and ΔU is equal to :

(1) – 4RT

(2) 3RT

(3) 4RT

(4) –3RT

34) Pre-exponential factors of two different reactions of same order are identical. Let activation energy of first reaction exceeds the activation energy of second reaction by 20 kJ mol^{-1}. If k_1 and k_2 are the rate constants of first and second reaction respectively at 300 K, then $\ln \frac{k_2}{k_1}$ will be ..... (nearest integer) [R = 8.3 J K^{-1} mol^{-1}]

(1) 4

(2) 6

(3) 20

(4) 8

35) 200cc of xM potassium dichromate solution is required to oxidise 750cc of 0.6 M Mohr's salt solution in acidic medium.
then x is –

(1) 0.2

(2) 0.75

(3) 0.375

(4) 0.1875

36) At 298 K, the mole percentage of N_2(g) in air is 80%. Water is in equilibrium with air at a pressure of 10 atm. What is the mole fraction of N_2(g) in water at 298 K? (K_H for N_2 is 6.5×10^7 mm Hg)

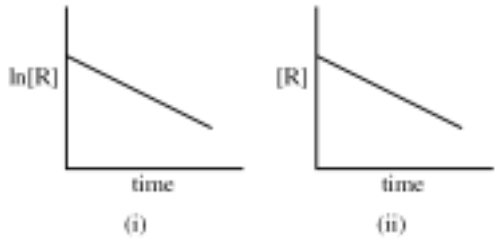
(1) 1.23×10^{-7}

(2) 1.17×10^{-4}

(3) 9.35×10^5

(4) 9.35×10^{-5}

37) The given plots represent the variation of the concentration of a reactant R with time for two different reactions (i) and (ii). The respective orders of the reaction are



(1) 1,1

(2) 0,1

(3) 1,0

(4) 0,2

38) What mass of $(NH_4)_2SO_4$ needs to be dissolved in 0.2 M ammonia solution to make one litre of buffer solution of pH = 8.26.
[Given $pK_b(NH_3)$ = 4.74, molar mass of NH_3 = 17 g mol^{-1}, molar mass of $(NH_4)_2SO_4$ = 132 g mol^{-1}]

(1) 12.4 g

(2) 24.8 g

(3) 132 g

(4) 248 g

39) For the given cell
$Pt(s)|H_2(g)|H^+(aq.)||Fe^{+3}(aq.), Fe^{+2}(aq.)|Pt(s)$

$E^0_{Fe^{+3}|Fe^{+2}} = 0.77V, \frac{2.303RT}{F} = 0.06$

$P_{H_2(g)} = 1atm, [H^+] = 1M, \frac{[Fe^{+3}]}{[Fe^{+2}]} = 0.1M$

then E_{cell} is –

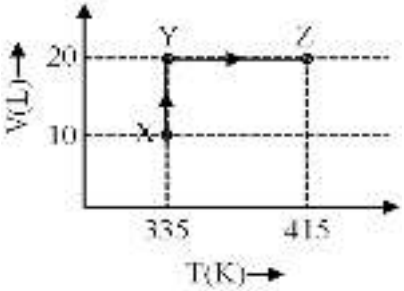
(1) 0.83 V

(2) 0.71 V

(3) – 0.23 V

(4) 1.77 V

40) Consider the following volume-temperature (V-T) diagram for the expansion of 5 moles of an ideal monoatomic gas.



Considering only P-V work is involved, the total change in enthalpy in kJ for the transformation of state in the sequence X → Y → Z is _____.
[Use the given data : Molar heat capacity of the gas for the given temperature range, $C_{V,m}$ = 12 J K^{-1}

mol^{-1} and gas constant, R = 8.3 J K^{-1} mol^{-1}]

(1) 4.8 kJ

(2) 2.4 kJ

(3) 8.120 kJ

(4) 16.4 kJ

41) Which of the following statement (s) is/are correct :-
(A) larger the value of K_a, stronger is the acid.
(B) Conjugate acid of H_2O is OH^-.
(C) pH of 10^{-8}M HCl(aq.) solution is 8.
(D) K_w of water increases with increase in temperature.
Choose the correct answer from the options given below :

(1) A, C, D

(2) A, D

(3) A, B,C, D

(4) A, B, D

42) Select the incorrect statement regarding equilibria involved in physical process :-

(1) For dissolution of solid in liquid, the solubility is constant at a given temperature.

(2) If some radioactive sugar is added in saturated solution of non-radioactive sugar, then after some time radioactivity is observed both in the solution and in the solid sugar.

(3) In case of liquid-vapour equilibrium, pressure exerted by vapour molecules at equilibrium becomes constant.

(4) When equilibrium is attained at a given temperature, the value of all its measurable properties are changing.

43) For same molal concentration, freezing point of which aqueous solution will be highest :-

(1) NaCl

(2) $C_6H_{12}O_6$

(3) $Ca_3(PO_4)_2$

(4) $MgCl_2$

44) An organic compound contains 10% hydrogen by mass, 0.01 mol of this compound on complete combustion produces 0.9 g H_2O, then molar mass of organic compound is –

(1) 100 g mol^{-1}

(2) 200 g mol^{-1}

(3) 150g mol^{-1}

(4) 50 g mol^{-1}

45) On the basis of following informations -

$$HCl\,(g) + 25aq \rightarrow HCl.25aq$$

ΔH = – 72 .03 kJ mol^{-1}

$HCl\ (g) + 40\ aq. \rightarrow HCl.40aq$
$\Delta H = -72.79\ kJ\ mol^{-1}$
enthalpy of dilution of HCl. 25 aq to HCl. 40 aq. is

(1) $-0.76\ kJ\ mol^{-1}$

(2) $+0.76\ kJ\ mol^{-1}$

(3) $-72.03\ kJ\ mol^{-1}$

(4) $-72.79\ kJ\ mol^{-1}$

BIOLOGY

1) Read the following statements regarding sex determination :
(a) In fact, the cytological observations made in a number of insects led to the development of the concept of genetic / chromosomal basis of sex-determination.
(b) X body of Henking was in fact a chromosome.
(c) In a large number of insects the mechanism of sex determination is of the XY type.
(d) Grasshopper is an example of XO type of sex determination.
(e) In human beings and in Drosophila the males have one X and one Y chromosome, whereas females have a pair of X-chromosomes besides autosomes.
Choose the correct statements from the options given below :

(1) b, d, e only

(2) a, c only

(3) a, c, e only

(4) a, b, d, e only

2) Given below are two statements :
Statement-I :- Morgan observed that the two genes segregate independently of each other.
Statement-II :- Morgan found that even genes were grouped on same chromosome, some genes were very tightly linked while others were loosely linked.
In the light of the above statements, choose the most appropriate answers from the options given below :

(1) Both statement I & statement II are correct

(2) Both statement I & statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

3) Match List-I with List-II

	List-I		**List-II**
(a)	Flower colour in the dog flower	(i)	Polygenic Inhertiance
(b)	AB blood group	(ii)	Incomplete dominance
(c)	Human skin colour	(iii)	Co-dominance
(d)	Phenylketonuria	(iv)	Pleiotropy

Choose the correct answer from the options given below :

(1) a-ii, b-iii, c-iv, d-i

(2) a-ii, b-iii, c-i, d-iv

(3) a-iii, b-ii, c-i, d-iv

(4) a-iv, b-iii, c-i, d-ii

4) If a karyotype of 47, XXY is present in an individual then which of the following is observed ?

(1) Overall masculine development

(2) Gynaecomastia

(3) Individual is sterile

(4) All of the above

5) How many different type of genotypes can be formed, in F_1 generation of a dihybrid cross ?

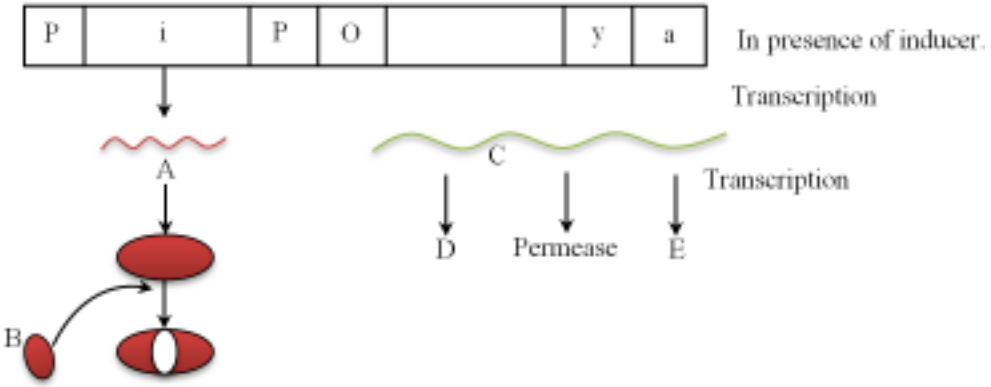
(1) 9

(2) 1

(3) 3

(4) 4

6) Given below digramatic sketch identify the parts labelled A, B, C and D, E and select the right option about them



(1) A-Inducer, B-Transacetylase, C-Repressor m-RNA, D-Lac-m-RNA, E-β-golactosidose

(2) A-Transacetylase, B-Repressor m-RNA, C-Lac m-RNA, D-Inducer, E-β-golactosidose

(3) A-Repressor m-RNA, B-Inducer, C-Lac mRNA, D-β-galactosidase, E-Transacetylase

(4) A-Lac-mRNA, B-Inducer, C-β-golactosidose, D-Transacetylase, E-repressor m-RNA.

7) Given below are two statements : one is labelled as assertion (A) and other is labelled as reason (R) :

Assertion (A) : In a transcription unit by switching the position of promoter with terminator, the definition of coding and template strands could be reversed.

Reason (R) : The presence of promoter in a transcription unit defines the template and coding strands.

In the light of above statements, choose the **correct** answer from the options given below :-

(1) Both A and R are true and R is the correct explanation of A.

(2) Both A and R are true but R is not the correct explanation of A.

(3) A is true but R is false.

(4) A is false but R is true.

8) Match List-I and List-II

	List-I		List-II
(a)	Pitch of DNA	(i)	0.34 nm
(b)	Distance between two base pair	(ii)	3.4 nm
(c)	Diameter of DNA	(iii)	36°
(d)	Each step of ascent	(iv)	2 nm

(1) (a)-(iii), (b)-(ii), (c)-(iv), (d)-(i)

(2) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)

(3) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

(4) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)

9) Which of the following gives the ability to DNA and RNA to direct their duplications ?

(1) Rule of base pairing and complementarity

(2) Ability of mutation

(3) Chemical and structural stability

(4) Expression of characters

10) The three codons which result in the termination of polypeptide chain synthesis are.

(1) UAA, UAG, UGA

(2) UAA, UAG, GUA

(3) UGU, UAG, UGA

(4) UAA, UGA, UUA

11) Scientists have identified about_____locations where single base DNA differences (SNPs-single nucleotide polymorphism, pronounced as snips') occur in humans.

(1) 3164.7 million

(2) 1,40.000

(3) 1.4 million

(4) 4.1 million

12) In photosystem-II the reaction centre chlorophyll a absorbs 680 nm wavelength of red light causing electrons to become excited and jump into an orbit farther from the atomic nucleus. These electrons are picked up by an electron acceptor which passes them to :-

(1) An electron transport system

(2) Oxysome

(3) Stroma of chloroplast

(4) Mobile carriers eg. UQ and cytochrome 'C'

13) **Assertion (A) :** Despite the presence of a green leaf, optimal light and CO_2 conditions, the plant may not photosynthesis, if the temperature is very low.
Reason (R) : When a chemical process is affected by more than one factor; then its rate will be determined by the factor which is nearest to its minimal value.
Select the correct option.

(1) **(A)** is not correct but **(R)** is correct.

(2) **(A)** is correct but **(R)** is not correct.

(3) Both **(A)** and **(R)** are correct and **(R)** is not the correct explanation of **(A)**.

(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**.

14)

	Column I (Stage of Respiration)		**Column II** (Primary Location in Eukaryotic Cell)		**Column III** (Major Output per Glucose Molecule)
(A)	Glycolysis	(i)	Mitochondrial Matrix	(1)	2 ATP (net), 2 NADH, 2 pyruvate
(B)	Pyruvate Oxidation (Link Reaction)	(ii)	Cytosol	(2)	2 Acetyl-CoA, 2 NADH, $2CO_2$
(C)	Citric Acid Cycle (Krebs cycle)	(iii)	Inner mitochondrial membrane	(3)	6 NADH, 2 $FADH_2$, 2ATP (GTP), $4CO_2$
(D)	Electron Transport Chain & Oxidative Phosphorylation	(iv)	Intermembrane space	(4)	~34 ATP, H_2O

(1) A-(i)-(1); B-(ii)-(2); C-(ii)-(3); D-(iii)-(4)

(2) A-(ii)-(1); B-(i)-(2); C-(i)-(3); D-(iii)/(iv)-(4)

(3) A-(i)-(1); B-(iii)-(3); C-(ii)-(2); D-(iv)-(4)

(4) A-(iv)-(4); B-(ii)-(2); C-(i)-(3); D-(iii)-(1)

15) Which statement is correct ?

(1) The passage of protons through the F_0 channel is coupled to the catalytic site of the F_1 component for the production of ATP.

(2) During the conversion of succinyl-CoA to succinic acid a molecule of ADP is synthesized.

(3) In cytoplasm, glucose has been broken down to release CO_2 and 8 molecules of NADH + H^+ ; two of $FADH_2$ have been synthesized

(4) The metabolic pathway through which the electrons passes from one carrier to another, is called the ETS and is present in matrix of mitochondria.

16) In Respiratory process which step is not related with de-carboxylation

(1) Alcoholic fermentation

(2) TCA cycle

(3) Oxidative decarboxylation of pyruvate

(4) Lactic acid Fermentation

17) Match the column-I with column-II & select the correct option.

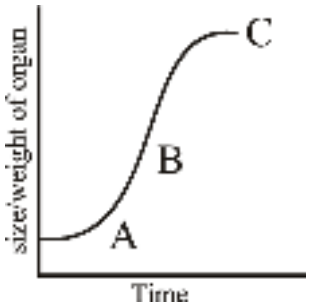
	Column-I		**Column-II**
a.	Stress hormone	i.	Auxin
b.	Internode elongation	ii.	Ethylene
c.	Gaseous PGR	iii.	Gibberellins
d.	Apical dominance	iv.	ABA

(1) a– iv, b–iii, c – ii, d–i

(2) a– ii, b–iv, c – i, d–iii

(3) a– i, b–ii, c – iii, d–iv

(4) a– iii, b–i, c – iv, d–ii

18) 

Choose the correct options for A, B, C given in above figure respectively :-

(1) A-Log phase, B-Lag phase, C-Stationary phase

(2) A-Lag phase, B-Log phase, C-Stationary phase

(3) A-Stationary phase, B-Lag phase, C-Log phase

(4) A-Lag phase, B-Stationary phase, C-Log phase

19) ES and EP complex are

(1) Both are unstable

(2) Both are stable

(3) ES complex is unstable while EP complex is stable

(4) EP complex is unstable while ES complex is stable

20) **Assertion:** Competitive inhibitors increase the activity of enzymes
Reason: Competitive inhibitors bind to the substrate binding site of the enzyme.

(1) Both assertion and reason are true and reason is the correct explaination of assertion.

(2) Both assertion and reason are true, but reason is not the correct explainaton of assertion.

(3) Assertion is true, but reason is false.

(4) Assertion false but reason is true

21) Match the column I with column II :

	Column-I		Column-II
(A)	Lectins	(I)	Protein
(B)	Inulin	(II)	Concanavalin-A
(C)	Insulin	(III)	Lipid
(D)	Lecithin	(IV)	Polysaccharide

Choose the correct answer from the options given below :

(1) A-II, B-I, C-IV, D-III

(2) A-IV, B-II, C-I, D-III

(3) A-II, B-IV, C-I, D-III

(4) A-III, B-IV, C-I, D-II

22) How many of the following are secondary metabolites :
Abrin, Ricin, Cellulose, Rubber, morphine, codeine, glycogen, starch.

(1) 5

(2) 6

(3) 7

(4) 8

23) As both types of gametes are non- motile in angiosperms, they have to be brought together for fertilization. This is achieved by pollination
Two statements A and B regarding pollination are given below.
Statement A: Cross pollination can be occurred by biotic and abiotic agent
Statement B: Cross pollination by biotic agent is more common in angiosperms and mainly occurs by insect

(1) Statement A is true but B is false

(2) Statement A and B both are true

(3) Statement B is true but A is false

(4) Statement A and B both are false

24) Which of the following is incorrect statements :-

(1) Endosperm development always begins after embryo formation.

(2) Coconut water is an example of free-nuclear endosperm

(3) Scutellum in grasses is a modified cotyledon

(4) Plumule is the part of embryo that develops into shoot.

25) Which of the following work is not associated with Dr. Panchanan Maheshwari ?

(1) Test tube fertilization in flowering plants

(2) Intra ovarian pollination

(3) Use of embryological characters in taxonomy

(4) He was awarded the triple crown of biology.

26) Which one of the following statement is not correct with regarding to the structure of dicot roots.

(1) The outermost layer is epiblema.
(2) The cortex consists of several layer of sclerenchyma cells with intercellcular spaces.
(3) Endodermis comprises a single layer of barrel shaped cells without any intercellular spaces.
(4) The pith is small or inconspicuous.

27) Select the option which is more appropriate for identification of given anatomical structure that



ensure the given anatomical structure is dicotyledmons stem :-

(1) Presence of selerenchymatons hypodmermis
(2) Vascular bundles are scattered closed
(3) Vascular bundles are arranged in a ring.
(4) Lack of well demarcated in a ring.

28) Read the following statements and choose the set of correct statements :
(A) The arrangement of veins and the veinlets in the lamina of leaf is termed as phyllotaxy.
(B) Leaves originate from shoot apical meristems.
(C) Leaves develops at the internode and bears a bud in its axil.
(D) A bud is present in the axil of petiole in both simple and compound leaves.
(E) In palmately compound leaves, the leaflets are attached at the tip of petiole.
Choose the **correct** answer from the options given below :

(1) B, D, E only
(2) A, C, D only
(3) C, D, E only
(4) A, B, C only

29) Match column-I with column-II.

	Column-I		**Column-II**
I.	Valvate	(A)	Pea
II.	Twisted	(B)	*Calotropis*
III.	Imbricate	(C)	Chinarose
IV.	Vexillary	(D)	*Cassia*

(1) I–B, II–C, III–D, IV–A

(2) I–A, II–B, III–C, IV–D

(3) I–D, II–C, III–B, IV–A

(4) I–A, II–C, III–A, IV–D

30) Read all statement carefully and select incorrect statement regarding to solanaceae family :-

(1) Stem branched , Solid or hollow, hairy or glabrous

(2) Persistent calyx, Valvate aestivation.

(3) Epipetalous stamens, Ovary superior

(4) Placenta swollen with one ovule and non-endospermic seed.

31) Arrange the following stages of the life cycle of gymnosperms in correct sequence and select correct answer.
(A) Seed germination
(B) Formation of strobili
(C) Pollination
(D) Discharge of male gametes into archegonia
(E) Formation of endosperm

Options :-

(1) B, E, C, A, D

(2) A, C, B, D, E

(3) B, C, E, D, A

(4) A, B, D, E, C

32) **Statement-I :** In pteridophytes, the main plant body is a gametophyte which is differentiated into true roots, stem and leaves.
Statement-II : The leaves in pteridophytes are small as in *Selaginella* or large as in fernes.

(1) Both statement-I and statement-II are correct.

(2) Both statement-I and statement-II are incorrect.

(3) Statement-I is correct and statement-II is incorrect.

(4) Statement-I is incorrect and statement-II is correct.

33) Match the column-I and column-II

	Column-I		**Column-II**
A.	Presence of algin	(i)	*Spirogyra*
B.	Presence of filamentous unbranched body	(ii)	*Dictyota*
C.	Presence of floridean starch	(iii)	*Porphyra*

(1) A-ii, B-i, C-iii

(2) A-i, B-ii, C-iii

(3) A-ii, B-iii, C-i

(4) A-iii, B-i, C-ii

34) A taxonomist is researching a group of fungi in a forest ecosystem. He observes that most of the fungi under study reproduce only by asexual sporse called conidia, and only their asexual or vegetative stages are known. The mycelium is septate and branched, and thse fungi are found growing on decaying leaves and wood, helping in decomposition and mineral cycling. The taxonomits initially classifies them as deuteromycetes (imperfect fungi). Later, while studying another region, he discovers a sexual reproductive stage in some of these fungi and identifies them as members of Ascomycetes.
Based on the above observations, which of the following conclusions is most accurate ?

(1) All deutermonycetes have both sexual and asexual stages, and they are never reclassified.

(2) Deuteromycetes are called 'imperfect fungi' because only their asexual or vegetative phases are known ; when their sexual stages are discovered, they are often reclassified into other classes like Ascomycetes or Basidiomycetes.

(3) Deuteromycetes are always found in aquatic enviroments, and their classification never changes.

(4) Deuteromycetes do not participate in decomposition or mineral cycling in any ecosystem.

35) Match the column-I with column-II and select the correct option from option given below :

Column-I (Disease)		**Column-II (Pathogen)**	
A.	Citrus canker	P.	Protozoa
B.	Malaria	Q.	Fungi
C.	Rust	R.	Viroid
D	Bovine spongiform encephalopathy	S.	Bacteria
		T.	Prions

Options:

(1) A:S, B:P, C:Q, D:T

(2) A:S, B:P, C:Q, D:R

(3) A:Q, B:P, C:S, D:T

(4) A:Q, B:P, C:S, D:R

36) According to five kingdom classification multicellular, heterotrophic cell walled organism belong to?

(1) Fungi kingdom only

(2) Fungi + Monera

(3) Fungi + Animalia

(4) Fungi + Monera + Animalia

37) which of the following statements are correct ?

(i) Species is basic unit of classification in taxonomy.
(ii) There is no suffix for genus, species & kingdom.
(iii) Domain is a super kingdom category.
(iv) Order is placed between division and class in taxonomic hierarchy.
(v) The earliest classification organisms where based on evolutionary relationship in organisms.

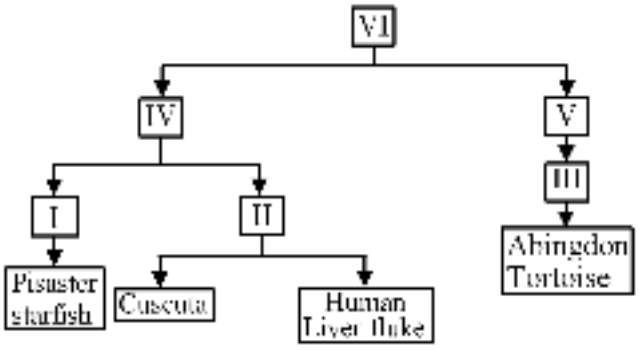
(1) i, ii, iii and iv

(2) i, ii and iii

(3) iii, iv and v

(4) i, iv and v

38) Recognise the following flow diagram and find correct option for population interactions :



	(I)	(II)	(III)	(IV)	(V)	(VI)
1.	Parasitism	Predation	Competition	–,–	–, 0	Neutral
2.	Predation	Parasitism	Competitive exclusion	+,–	–,–	Detrimental
3.	Predation	Parasitism	Competition	+, –	–,–	Neutral
4.	Competition	Commensalism	competitive release	–,0	+,0	Detrimental

(1) 1

(2) 2

(3) 3

(4) 4

39) Which of the following statement is correct ?

(1) Both species are harmed in mutualism.

(2) Both species are benefited in commensalism.

(3) One species is benefited and other is unaffected in commensalism.

(4) An orchid growing as an epiphyte on a mango branch is an example of mutualism.

40) In the logistic growth equation $dN/dt = rN\left[\frac{K-N}{K}\right]$, 'r' represents :

(1) Extrinsic rate of natural increase

(2) The base of rate of natural increase

(3) Intrinsic rate of natural increase

(4) Carrying capacity

41) Select the correctly matched column :

Column-(I)		Column-(II)		Column-(III)	
(A)	Pyramid of number	(I)	Inverted	(a)	Grassland
(B)	Pyramid of biomass	(II)	Low biomass	(b)	Aquatic ecosystem
(C)	Pyramid of energy	(III)	Upright	(c)	All natural ecosystems
(D)	Small standing crop	(IV)	Always upright	(d)	Phytoplankton in aquatic ecosystem

(1) A-III-b, B-II-d, C-IV-c, D-I-a

(2) A-III-a, B-I-b, C-IV-c, D-II-d

(3) A-II-a, B-IV-b, C-I-c, D-III-d

(4) A-I-a, B-III-c, A-II-b, C-I-c

42) Which one of the following is structural characteristics of ecosystem ?

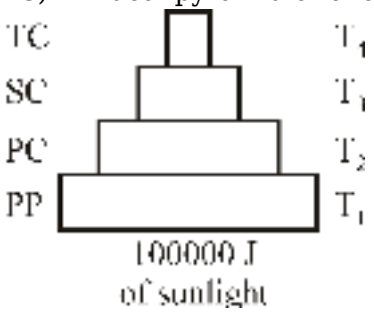
(1) Stratification

(2) Species dominancy

(3) Species diversity

(4) All of these

43) An ideal pyramid of energy given below. Find out the energy at tertiary consumer level :-



(1) 1 J

(2) 10 J

(3) 100 J

(4) 1000 J

44) **Statement-I :** Alien species invasion is the most important cause driving animals and plant to extinction.
Statement-II : Species extinctions in the last 500 years (Steller's Sea cow) were due to Habitat loss and fragmentation.

(1) Statement I and II both are correct

(2) Statement I and II both are incorrect

(3) Only Statement I is correct

(4) Only Statement II is correct

45) In "rivet popper hypothesis" (Paul Ehrlich) rivet indicates :-

(1) Ecosystem

(2) Key stone species

(3) Community

(4) Species

46) **Assertion (A) :-** In Molluscs, the body is unsegmented with a distinct head, muscular foot, and visceral hump.
Reason (R) :- A soft and spongy layer of skin forms a mantle over the visceral hump.

(1) Both A and R are true and R is the correct explanation of A

(2) Both A and R are true but R is not the correct explanation of A

(3) A is true but R is false

(4) A is false but R is true

47) Which of the following combinations is incorrect ?
1. Crocodilus – Four-chambered heart
2. Balaenoptera – Viviparous mammal
3. Aptenodytes – Flightless bird
4. Ornithorhynchus – Viviparous mammal

(1) 1 only

(2) 2 only

(3) 4 only

(4) 1 and 4

48) Match the **Column I** & **Column II** :-

	Column I		**Column II**
(a)	Chelone	(i)	Tortoise
(b)	Columba	(ii)	Vulture
(c)	Neophron	(iii)	Pigeon
(d)	Testudo	(iv)	Turtle

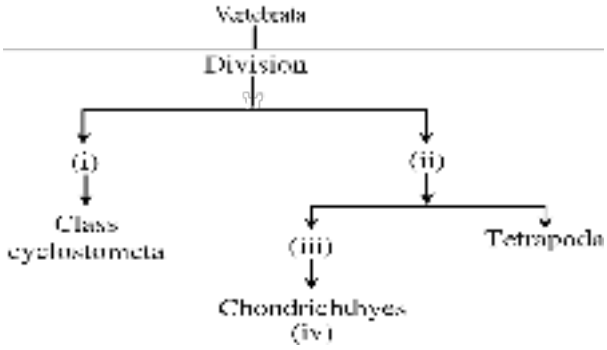
(1) a-i, b-ii, c-iii, d-iv

(2) a-iv, b-iii, c-ii, d-i

(3) a-iii, b-iv, c-i, d-ii

(4) a-ii, b-i, c-iii, d-iv

49) Observe the following chart carefully & fill the boxes ( ) with appropriate word choose the



correct option.

	i	ii	iii	iv
1	Agnatha	Gnathostomata	Pisces	Osteichthyes
2	Gnathostometa	Agnatha	Pisces	Reptilia
3	Cyclostometa	urochordate	Aves	Amphibia
4	Urochordates	Amphibia	Aves	Reptilia

(1) 1

(2) 2

(3) 3

(4) 4

50) The epithelium tissue has the following features :-
(A) Compound epithelium is multilayered having limited role in secretion and absorption.
(B) Forms tip of nose.
(C) Exocrine glands posses epithelium tissue.
(D) Bear cilia on their basal surface.
(E) Forms a lining for some parts of the body.
Choose the correct option :

(1) A, C, E

(2) A, D, E

(3) B, C, E

(4) C, D, E

51) Given below are two statements about cockroach :
Statement-I : In both sexes, the 10th segment bears a pair of jointed filamentous structures called anal style.
Statement-II : Males bear a pair of short, thread-like anal cerci which are absent in females.
In the light of the above statements, choose the most appropriate answer :

(1) Both Statement-I and Statement-II are incorrect.

(2) Statement-I is correct but Statement-II is incorrect.

(3) Statement-I is incorrect but Statement-II is correct.

(4) Both Statement-I and Statement-II are correct.

52) Some structures present in frogs are given below:
(A) Vocal sacs

(B) Tympanum
(C) Copulatory pads
(D) Webbed toes
(E) Bidder's Canal
Which of the above structures form the basis of sexual dimorphism in frogs?

(1) (B) and (C) only
(2) (A) and (B) only
(3) (A) and (C) only
(4) (A),(C) and (E) only

53) Which organ receives blood through the hepatic portal vein?

(1) Heart
(2) Kidney
(3) Liver
(4) Brain

54) Given below are two statements :
Statement-I : ECG is a graphical representation of the electrical activity of the heart during a cardiac cycle.
Statement-II : T-wave in ECG represents the return of the ventricles from excited to normal state.

(1) Both statement-I and statement-II are correct.
(2) Statement-I is correct but statement-II is incorrect.
(3) Statement-I is incorrect but statement-II is correct.
(4) Both statement-I and statement-II are incorrect.

55) Read the following (A - D) Statements :-
(A) Plasma is a straw coloured, viscous fluid constituting 55 percent of the blood.
(B) 90-92 percent of plasma is water and proteins contribute 6-8 percent of it
(C) Globulins are needed for clotting or coagulation of Blood
(D) Fibrinogens are primarily involved in defence mechanism of the body.
How many of the above statements are correct:

(1) Four
(2) Three
(3) Two
(4) One

56) Match the column-I and II.

	Column-I		**Column-II**
i.	Tidal volume (TV)	a.	1100-1200 ml
ii.	Inspiratory reserve volume (IRV)	b.	4800 ml

iii.	Residual volume (RV)	c.	500 ml
iv.	Vital capacity	d.	2500-3000 ml

Choose appropriate option :-

(1) i-a, ii-b, iii-c, iv-d

(2) i-c, ii-d, iii-a, iv-b

(3) i-c, ii-b, iii-d, iv-a

(4) i-a, ii-c, iii-d, iv-b

57) Which of the following structures are sensitive for CO_2 and H^+ concentration are -

A. Chemosensitive area
B. Chemoreceptor associated with aortic arch
C. Chemoreceptor associated with carotid artery
D. Carotid body
E. Aortic body

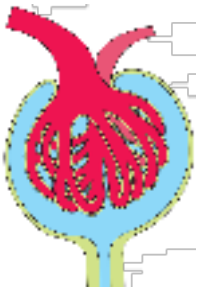
(1) Only A, B, E

(2) Only B, C, D

(3) Only A, C, E

(4) All A, B, C, D and E

58) Renal tubule begins with -



(1) Single walled cup shaped capillaries.

(2) Double walled cup shaped glomerulus.

(3) Double walled cup shaped structure.

(4) Single walled cup shaped structure.

59) **Assertion (A) :** In LOH, when filtrate move down in descending LOH, it gets hyperosmotic, when it move upwards in ascending LOH, it gets hypoosmotic to aid more electrolyte in medullary interstitum.
Reason (R) : Descending LOH is impermeable to electrolyte and permeable to water, where as ascending LOH is impermeable to water and allow transport of electrolyte actively or passively.

(1) Both (A) & (R) are true and (R) is correct explanation of (A)

(2) Both (A) & (R) are true and (R) is not correct explanation of (A)

(3) Both (A) and (R) are false

(4) (A) is true but (R) is false

60) The axoplasm contains high concentration of ___(A)___ and negatively charged proteins and low

concentration of ____(B)____ ?

	(A)	(B)
(1)	$K^{\oplus}$ ions	$PO_4^{\ominus}$ ions
(2)	$Na^{\oplus}$ ions	$K^{\oplus}$ ions
(3)	$K^{\oplus}$ ions	$Na^{\oplus}$ ions
(4)	PO_4^{-3} ions	$K^{\oplus}$ ions

(1) 1

(2) 2

(3) 3

(4) 4

61)

Column-I		**Column-II**	
(A)	Multipolar	(i)	Retina of eye
(B)	Bipolar	(ii)	Cerebral Cortex
(C)	Unipolar	(iii)	Cell body
(D)	Nissl's granules	(iv)	Embryonic stage

Codes :

	A	**B**	**C**	**D**
(1)	ii	iii	i	iv
(2)	ii	i	iii	iv
(3)	ii	iv	i	iii
(4)	ii	i	iv	iii

(1) 1

(2) 2

(3) 3

(4) 4

62) Which one of following hormone promote given process (P–T) –
P. Gluconeogenesis
Q. Lipolysis
R. Proteolysis
S. Glycogenolysis
T. Inhibit utilisation of amino acids

(1) Hormone of zona fasciculata and α-cells of pancreas

(2) Hormone of zona reticularis and α-cells of pancreas

(3) Hormone of zona fasciculata and β-cells of pancreas

(4) Hormone of zona reticularis and β-cells of pancreas

63) (a) Pineal secretes a hormone called melanin.
(b) Deficiency of iodine causes goitre.
(c) Deficiency of ADH causes type-I diabetes
(d) PTH is a hypercalcemic hormone
Read the above statements and choose the correct option:

(1) a & b

(2) c & d

(3) b & d

(4) a & d

64) Choose the correct answer from the given below options about hormone action mechanism.

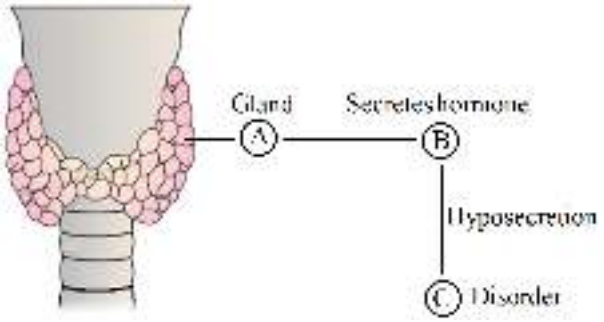
(1) Steroidal hormone do not enter the target cell but intract with cell membrane only.

(2) Hormone which interacts with intra cellular receptors are proteinaceous hormone.

(3) Protein nature hormone generates the secondary messanger after binding with cell membrane.

(4) Biochemical response is directly stimulated by protein hormones in cytoplasm of target cell.

65)



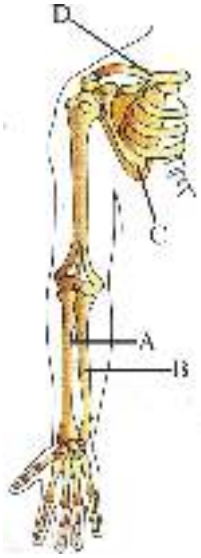
Choose the correct option for (A), (B) & (C) labelling :-

(1) A – Parathyroid, B – PTH, C – Tetany

(2) A – Thyroid, B – T_4, C – Cretinism

(3) A – Thyroid, B – T_4, C – Gigantism

(4) A – Parathyroid, B – PTH, C – Hypocalcemia



66) In given diagram A,B,C and D respectively represents :-

(1) Radius, Ulna, Scapula, Clavicle

(2) Ulna, Radius, Scapula, Clavicle

(3) Radius, Ulna, Clavicle, Scapula

(4) Ulna, Radius, Clavicle, Scapula

67) Given below are two statements:
Statement I: Each 'F' (Filamentous) actin is a polymer of monomeric 'G'(Globular) actin.
Statement II: Each myosin filament is a polymer of monomeric proteins called tropomyosins.
In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement I and Statement II are incorrect.

(2) Statement I is correct but Statement II is incorrect.

(3) Statement I is incorrect but Statement II is correct.

(4) Both Statement I and Statement II are correct.

68) Early embryo upto 8 blastomeres formed due to in vitro fertilization is transferred into :-

(1) Fimbriae

(2) Cervix

(3) Uterus

(4) Fallopian tube

69) Given below are two statements one is labelled as Assertion (A) and another is labelled as reason (R).
Assertion (A) :- Placenta is a structural and functional unit between developing embryo and maternal body formed by the interdigitation between chorionic villi and uterine tissue.
Reason (R) :- Placenta acts as endocrine tissue and produces several hormones.
Select the correct answer from the options given below :

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(2) Both (A) and (R) are correct but (R) is not correct explanation of (A)

(3) (A) is correct but (R) is incorrect

(4) (A) is incorrect but (R) is correct

70) Match the List-I with List-II and choose the correct match from the options given below :

	List-I		**List-II**
(A)	Ampulla	(I)	Acts on sertoli cells
(B)	Hymen	(II)	Finger-like projections
(C)	FSH	(III)	Wider part of fallopian tube
(D)	Chorionic villi	(IV)	Not a reliable indicator of virginity

(1) A-III ; B-I ; C-IV ; D-II

(2) A-III; B-IV ; C-I ; D-II

(3) A-IV ; B-III ; C-I ; D-II

(4) A-I ; B-IV ; C-II ; D-III

71) Read the following two statements and choose the correct option.
Statement-I : Each testis has about 250 compartments called testicular lobules.
Statement-II : Leydig cells synthesise and secrete tesicular hormones called androgens.

(1) Both Statement I and Statement II are incorrect.

(2) Both Statement I and Statement II are correct.

(3) Only Statement I is correct.

(4) Only Statement II is correct.

72) Read the following paragraph:
The primary sex organs – the testis in the males and the ovaries in the females –produce gametes, i.e, sperms and ovum, respectively, by the process called gametogenesis. There are remarkable differences between the reproductive events in the male and in the female, for example, sperm formation continues even in old men, but formation of ovum ceases in women around the age of fifty years.
Given below are two statements about this:
Statement I: The spermatogonia present on the inside wall of seminiferous tubules multiply by mitotic division and increase in numbers. Each spermatogonium is haploid and contains 23 chromosomes.
Statement II: The secondary follicle soon transforms into a tertiary follicle which is characterised by a fluid filled cavity called antrum. The theca layer is organised into an inner theca interna and an outer theca externa.
In the light of above statements, choose the most appropriate answer from the options given below:

(1) Both the statements are correct

(2) Both the statements are incorrect

(3) Only statement I is correct

(4) Only statement II is correct

73) Identify the contraceptive device shown below as well as the related right site of its implantation



into a woman. Then, select the option with correct matching.

	Contraceptive device	**Site of implant**
(1)	LNG – 20	Fallopian tube
(2)	Lippes loop	Uterine wall
(3)	Implant	Subcutaneous tissue
(4)	Multiload 375	Uterine wall

(1) 1

(2) 2

(3) 3

(4) 4

74) Which of the following is not an autoimmune disease?

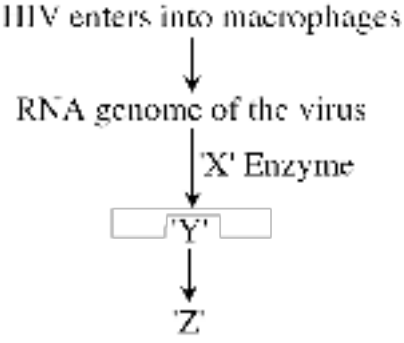
(1) Allergy

(2) Rheumatoid Arthritis

(3) Hashimoto's disease

(4) Myasthenia gravis

75) After getting into the body of the person, the HIV enters into macrophages where RNA genome of the virus replicates to form Y with the help of the 'X' enzyme. Now 'Y' gets incorporated into 'Z' and directs the infected cells to produce virus particles. The macrophages continue to produce virus and in this way acts like a HIV factory. The flow chart given below shows the replication of HIV



Which of the following correct regarding 'X', Y and Z.

(1) X = Protease enzyme; Y-Viral DNA; Z = Host cells DNA

(2) X = Ligase enzyme; Y = Host cells DNA; Z = Viral DNA

(3) X = Reverse transcriptase enzyme; Y = Viral DNA; Z = Host cells DNA

(4) X = RNA polymerase enzyme; Y = Viral DNA; Z = Host cells DNA

76) Given below are two statements:
Statement-I :- The filarial worms cause a slowly developing chronic inflammation of the organs in

which they live for many years.
Statement-II :- Muscular pain, fever, internal bleeding, anemia and blockage of the intestinal passage are the common symptoms of typhoid fever.
In the light of the above statement choose the correct answer from the options given below:

(1) Statement-I and II both are correct

(2) Statement-I and II both are incorrect

(3) Statement-I is correct but II is incorrect

(4) Statement-I is incorrect but II is correct

77) Read the following statements :-
(A) Skin on our body is the main barrier which allow entry of the micro-organisms.
(B) Acid in stomach and saliva in mouth prevent microbial growth
(C) Innate immunity is non-specific type of deffence.
(D) Innate immunity is characterised by memory.
How many of the above statements are incorrect?

(1) Two

(2) Three

(3) One

(4) Four

78) Read both the given statements on industrial melanism and select the correct option.
Statement-I: Before industrialisation set in, the growth of almost white-coloured lichen covered the trees in that background the dark winged moth survived but the white coloured moth were picked out by predators.
Statement-II: During post-industrialisation period, the tree trunks became dark due to industrial smoke and soots. Under this condition the white-winged moth survived.

(1) Both statements are correct

(2) Both statements are incorrect

(3) Only statement-I is correct

(4) Only statement-II is correct

79) Match the hominids with their approximate brain capacities:

	List-I		**List-II**
(A)	Homo habilis	(1)	1400 cc
(B)	Homo erectus	(2)	650-800 cc
(C)	Neanderthal man	(3)	900 cc

(1) A-2, B-3, C-1

(2) A-1, B-2, C-3

(3) A-3, B-2, C-1

(4) A-2, B-1, C-3

80) Select the correct statement about centrioles -

(1) Centriole is surrounded by a membrane

(2) Centriole help in chromosomal division in angiosperm

(3) Both the centrioles in a centrosome lie perpendicular to each other

(4) Central region of the centriole has two single microtubules

81) **Assertion:** Ribosomes are found in all cells i.e. in both eukaryotic as well as in prokaryotic cells.
Reason: Ribosomes are non membrane bound cell organelles.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

82) Given below are two statements :
Statement I : Each mitochondrion is a double membrane-bound structure with the outer membrane and the inner membrane dividing its stroma distinctly into two aqueous compartments.
Statement II : In the chromoplasts water soluble carotenoid pigments like carotene, xanthophylls and others are present.
In the light of the above statement, choose the most appropriate answer from the options given below:

(1) Both Statement I and Statement II are correct.

(2) Both Statement I and the Statement II are incorrect.

(3) Statement I is correct but Statement II is incorrect

(4) Statement I is incorrect but Statement II is correct.

83) Few events regarding meiosis is given below. Arrange these events in ascending order and select the correct option :-
(A) Segregatin of homologous chromosome
(B) Terminalisation of chiasmata.
(C) Arrangement of bivalent on equator.
(D) Formation of synaptonemal complex.
(E) Exchange of genetic material.
Option :-

(1) D → B → E → C → A

(2) D → E → B → C → A

(3) E → D → B → A → C

(4) E → B → D → C → A

84) Nucleolus:-

(1) Is covered by nucleolar membrane

(2) Is larger in cells involved in sterol synthesis

(3) Is found in nucleoplasm

(4) Encloses whole chromatin inside it

85) Which of the following is basic requirement of PCR reaction ?

(1) Two oligonucleotide primer

(2) DNA segment to be amplified

(3) A heat stable DNA polymerase

(4) All of the above

86) Select the correct option-
Statement - I : In a chromosome there is specific DNA sequence called origin of replication, which is responsible for initiating replication.
Statement - II : Plasmid DNA act as vectors to transfer the piece of DNA attached to it.

(1) Both statement are correct.

(2) Both statement are incorrect.

(3) Only statement - I is correct.

(4) Only statement - II is correct.

87) Observe the diagram given below showing a recombinant DNA experiment using the plasmid



pBR322. After transformation, bacteria are grown on two plates:
Plate A: Contains ampicillin
Plate B: Contains tetracycline
Bacteria that grow on Plate A but do not grow on Plate B are considered recombinant.
What conclusion can be drawn from the growth pattern of the bacteria?

(1) The inserted foreign DNA restored the function of the tet^R gene.

(2) The plasmid failed to incorporate the foreign gene.

(3) The tet^R gene was disrupted by insertion of the foreign DNA, indicating successful recombination.

(4) Bacteria growing on both plates contain recombinant plasmids

88) Match the column–I with column–II

Column -I		**Column - II**	
(i)	Golden rice	(A)	Increased shelf life
(ii)	Flavr savr tomato	(B)	HGH
(iii)	Mouse	(C)	Vitamin A

(iv)	*Brassica napus*	(D)	Hirudin

(1) (i) – C, (ii) – A, (iii) – B, (iv) – D

(2) (i) – B, (ii) – A, (iii) – C, (iv) – D

(3) (i) – B, (ii) – C, (iii) – A, (iv) – D

(4) (i) – D, (ii) – A, (iii) – B, (iv) – C

89) Consider the following statements-
I. Transgenic animals are more sensitive to the toxic substance than non transgenic animal.
II. Useful biological products can be produced by introducing the portion of DNA which codes for a particular product in to transgenic animals.
III. Transgenic animal are designed to allow the study of gene regulation.
IV. Transgenic animals are used to study of normal physiology and development.
Which of the above statements are correct.

(1) I, II, III, IV

(2) I, II

(3) I, III

(4) III, IV

90) Consider the following four statements (a-d) and select the option which includes all the correct ones only.
(a) Blue green algae also add organic matter to the soil and increase its fertility.
(b) *Trichoderma* species are free-living fungi that are very common in the root ecosystem.
(c) Many members of the genus *Rhizobium* form mycorrhiza.
(d) Microbes are used to produce industrial products like curd, cheese, idli and dosa.

(1) Statements (b) , (c) and (d)

(2) Statements (a) , (b)

(3) Statements (c) , (d)

(4) Statements (a) , (c) and (d)

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	3	1	4	4	4	1	4	3	1	1	1	1	1	1	2	2	2	1	4	4
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	2	3	4	1	3	2	2	1	1	4	3	4	2	2	1	1	2	3	3	2
Q.	41	42	43	44	45															
A.	4	1	3	2	4															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	3	3	4	3	1	3	2	1	2	4	2	4	4	3	4	1	1	2	4	2
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	3	4	1	3	4	2	1	1	2	3	2	4	1	4	3	4	3	3	2	3
Q.	86	87	88	89	90															
A.	2	4	2	1	1															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	4	4	2	4	2	3	1	3	1	1	3	1	4	2	1	4	1	2	1	4
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	3	2	2	1	4	2	3	1	1	4	3	4	1	2	1	1	2	2	3	3
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	4	1	2	4	2	3	2	1	1	1	3	3	1	3	2	4	3	1	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	4	1	3	3	2	1	2	4	2	2	2	4	3	1	3	3	1	2	1	3
Q.	171	172	173	174	175	176	177	178	179	180										
A.	2	2	2	3	4	1	3	1	1	2										

SOLUTIONS

PHYSICS

1) $t \propto T^x m^y \square^z$
$[T] = [MLT^{-2}]^x [ML^{-1}]^y [L]^z$
$[M^0L^0T^1] = [M^{x+y} L^{x-y+z} T^{-2x}]$
$\Rightarrow -2x = 1 \Rightarrow x = -1/2$
$\Rightarrow x + y = 0 \Rightarrow y = -x = 1/2$
$x - y + z = 0 \Rightarrow z = y - x = 1$
$t \propto T^{-1/2} m^{1/2} \square^1$

2) Both the projectiles will be projected at complimentary angles

$$t_1 = \frac{2u\sin\theta}{g}$$

$$t_2 = \frac{2u\sin(90^\circ - \theta)}{g} = \frac{2u\cos\theta}{g}$$

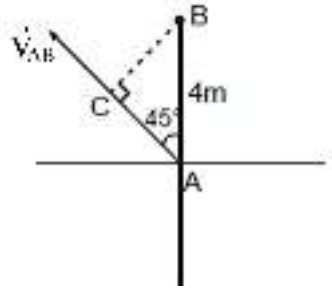
$$t_1 t_2 = \frac{2(u\sin\theta)(u\cos\theta) \times 2}{g \times g}$$

$$t_1 t_2 = \frac{2R}{g}$$

$$\Rightarrow R = \frac{gt_1 t_2}{2}$$

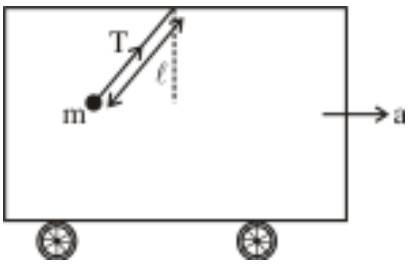
3) $\vec{v}_{AB} = \vec{v}_A - \vec{v}_B = 2\left(\hat{j} - \hat{i}\right)$ m/s or $|\vec{v}_{AB}| = 2\sqrt{2}$ m/s

Assuming B to be at rest, A will move with velocity $\vec{v}_{AB}$ in the direction shown in figure. The distance between them will first decrease from A to Cand then increase beyond C.



Minimum distance between them is BC which is equal to $\frac{4}{\sqrt{2}}$ or $2\sqrt{2}$ and the time after which they are at closest distance is :

$$t = \frac{AC}{|\vec{v}_{AB}|} = \frac{2\sqrt{2}}{2\sqrt{2}} = 1\text{s}$$

4) 

As ball is at rest with respect to trolley so that work done by tension force w.r.t. trolley is zero.

5)

During regulation of zener diode, it offer variable resistance, I_Z changes but V_Z remains constant and I_S (supply current) through the series resistance (R_S) changes.

6) $Y = \overline{A \cdot B} = \bar{A} + \bar{B} = A + B$

7)

In interference, the maximum intensity (I_{max}) occurs when amplitudes add, and the minimum intensity (I_{min}) occurs when amplitudes subtract. The intensity (I) is proportional to the square of the amplitude (A). We use the relation $\frac{I_{max}}{I_{min}} = \left(\frac{A_1 + A_2}{A_1 - A_2}\right)^2$, where A_1 and A_2 are the amplitudes of the interfering waves.
Given ratio of intensities:

$$\frac{I_{max}}{I_{min}} = \frac{36}{1}$$

Using the amplitude relation:

$$\left(\frac{A_1 + A_2}{A_1 - A_2}\right)^2 = \frac{36}{1}$$

Taking the square root of both sides:

$$\frac{A_1 + A_2}{A_1 - A_2} = \sqrt{\frac{36}{1}} = \frac{6}{1}$$

Using the property of componendo and dividendo:

$$\Rightarrow \frac{2A_1}{2A_2} = \frac{7}{5}$$

The ratio of amplitudes is: $\frac{A_1}{A_2} = \frac{7}{5}$

8)

Stopping potential depends upon frequency of incident light, not on light intensity which contradicts wave theory of light.

9) Q = 8.4 (139 + 100) – 7.5 × 239
= 239 [8.4 – 7.5]

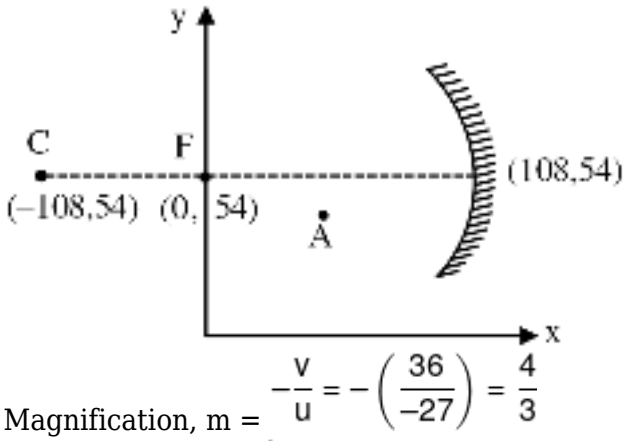
= 239 × 0.9
= 215.1 MeV

10) $\Delta m = |Z m_p + N m_n - m(^A_Z X)]$
=[3 × 1.007277 + 4 × 1.008665 – 7.016005] u
= 0.04048 u

11) $[IE]_Z = Z^2 (IE)_H$
= 9 × 13.6 eV

12) $|M| = \left|\left(\frac{f_0}{f_0 + u_0}\right)\left(1 + \frac{D}{f_e}\right)\right|$
$= \left|\left(\frac{1}{1-1.1}\right)\left(1 + \frac{25}{5}\right)\right| = 60$

13) Here, f = – 108 cm, u = – 27 cm
$\because \frac{1}{v} + \frac{1}{u} = \frac{1}{f}$
$\therefore \frac{1}{v} + \frac{1}{-27} = \frac{1}{-108} \Rightarrow v = 36$ cm
⇒ X–coordinate will be 36 + 108 = 144 cm



Magnification, m = $-\frac{v}{u} = -\left(\frac{36}{-27}\right) = \frac{4}{3}$
Height of image = $\frac{4}{3}$ × 6 = 8 cm
⇒ Y–coordinate = 54– 8 = 46 cm

14) $\delta = \left|\frac{\mu_p}{\mu_s} - 1\right| A$
$4 = \left|\frac{1.5}{1} - 1\right| A$ ... (1)
$\delta = \left|\frac{1.5}{\frac{4}{3}} - 1\right| A$ ....(2)
(1)/(2) ⇒ δ = 1°

15) $R = \frac{\sqrt{2mE}}{qB}$

$R \propto \frac{\sqrt{E}}{B}$

$R = \frac{\sqrt{2}}{3}R$

16)

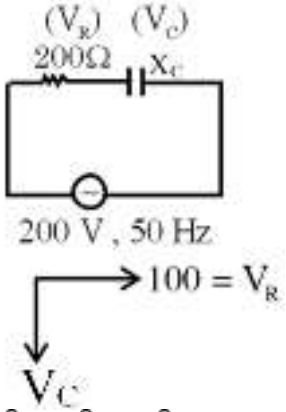
Ampere's law $\oint \vec{B}.d\vec{\ell} = \mu_o I_{enclosed}$

$r < R$, $I_{enclosed} = \frac{Ir^2}{R^2}$

$r \geq R$, $I_{enclosed} = I$

17) $P = \frac{V^2}{R} \Rightarrow R = \frac{V^2}{P}$

$R = \frac{100 \times 100}{50} \Rightarrow R = 200\Omega$



$V_R^2 + V_C^2 = V^2$

$(100)^2 + V_C^2 = (200)^2$

$\Rightarrow V_C = 100\sqrt{3}$ volts

$I = \frac{P}{V} = \frac{50}{100} = \frac{1}{2}A$

$V_C = I \times X_C$

$\Rightarrow X_C = 200\sqrt{3}$

$\Rightarrow 200\sqrt{3} = \frac{1}{\omega C}$

$C = \frac{1}{2\pi \times 50 \times 200\sqrt{3}} = \frac{50 \times 10^{-6}}{\pi\sqrt{x}}$

$\sqrt{x} = 50 \times 10^{-6} \times 100 \times 200\sqrt{3}$

$x = 3$

18)

EMW spectrum does not consists of clear & sharp division of different kind of waves.

19) $I = \frac{P}{A} = \frac{200 \times \frac{5}{100}}{4\pi(10)^2}$

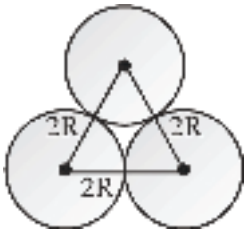
$= 0.008\ W/m^2$

20) By law of conservation of mechanical energy, we have

$(U+K)_{initial} = (U+K)_{final}$

$\Rightarrow -\frac{3Gm^2}{d} + 3\left(\frac{1}{2}m(0)^2\right) = -\frac{3Gm^2}{2R} + 3\left(\frac{1}{2}mv^2\right)$

$\Rightarrow 3\left(\frac{1}{2}mv^2\right) = 3\left(\frac{Gm^2}{2R} - \frac{Gm^2}{d}\right)$



$\Rightarrow v^2 = Gm\left(\frac{1}{R} - \frac{2}{d}\right)$

$\Rightarrow v = \sqrt{Gm\left(\frac{1}{R} - \frac{2}{d}\right)}$

21) $(4m)(U) + m(0) = (4m)v_1 + mv_2$

$4U = 4v_1 + v_2$ .... (1)

$\because e = \frac{v_2 - v_1}{u_1 - u_2} \Rightarrow 1 = \frac{v_2 - v_1}{U}$

$\Rightarrow v_2 - v_1 = U$ ... (2)

eq. (1) – eq. (2) $\Rightarrow 5v_1 = 3U \Rightarrow v_1 = \frac{3U}{5}$

fraction $\Rightarrow \frac{\frac{1}{2}(4m)U^2 - \frac{1}{2}(4m)v_1^2}{\frac{1}{2}(4m)U^2} = \frac{16}{25}$

22) Let r be the radius of each small droplets.

Then, volume of big drop = 64 × volume of each small droplets

$\frac{4}{3}\pi R^3 = 64 \times \frac{4}{3}\pi r^3$

$R = 4r$ ...(i)

Surface area of big drop = $4\pi R^2$

Surface area of 64 small droplets = $64 \times 4\pi r^2$

$\therefore$ Increase in surface area = $64 \times 4\pi r^2 - 4\pi R^2$

$= 4\pi[64r^2 - R^2]$

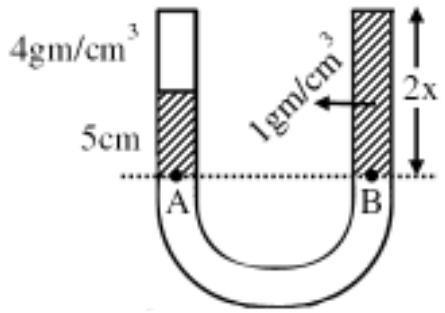
$= 4\pi[4R^2 - R^2]$ [Using eqn.(i)]

$= 12\pi R^2$

Energy needed = surface tension × Increase in surface area

$= T \times 12\pi R^2 = 12\pi R^2 T$

23) Let water in one arm is depressed by x



$P_A = P_B$

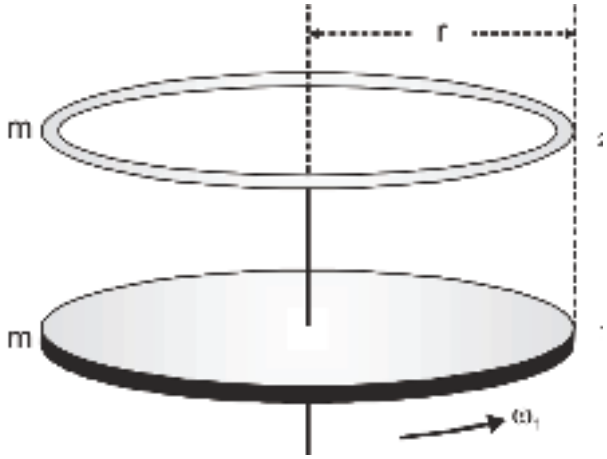
$4 \times g \times 5 = 1 \times g \times 2x$

$x = 10$

$h_1 = 30 - x + 5 = 30 - 10 + 5 = 25$ cm

$h_2 = 30 + x = 30 + 10 = 40$ cm

$$\frac{h_2}{h_1} = \frac{40}{25} = \frac{8}{5}$$



24)

Formula : $L_{initial} = L_{final}$

$$K = \frac{1}{2} I\omega^2$$

$$\because I_{disc} = \frac{1}{2} MR^2$$

$$: I_{ring} = MR^2$$

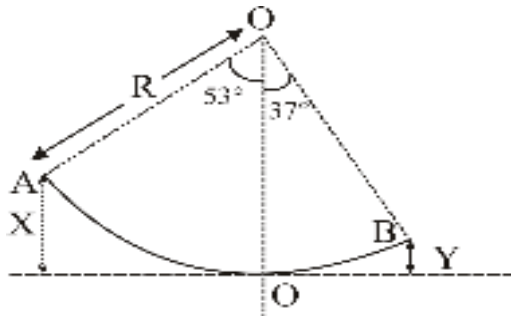
Heat produced = loss in kinetic energy = $E_1 - E_2$

Ratio heat produced to initial kinetic energy = $\frac{E_1 - E_2}{E_1}$

$$= 1 - \frac{E_2}{E_1} = 1 - \left(\frac{I_1 + I_2}{I_1}\right)\left(\frac{\omega_2}{\omega_1}\right)^2$$

$$= 1 - \left(1 + \frac{I_2}{I_1}\right)\left(\frac{\omega_2}{\omega_1}\right)^2$$

$$= 1 - (1 + 2)\left(\frac{1}{3}\right)^2 = 1 - \frac{1}{3} = \frac{2}{3}$$



25)

$X = R(1 - \cos 53°) = 0.4R$

$Y = R(1 - \cos 37°) = 0.2R$

Apply COME at A and B

$mgX = mgY + \frac{1}{2}mv^2$

$mg(0.4R) = mg(0.2R) + \frac{1}{2}mv^2$

$v^2 = 0.4gR$

Block leaves at B, $N = 0$

$\frac{mv^2}{r} = mg\cos 37°$

$m\frac{(0.4gR)}{r} = mg(0.8)$

$r = \frac{R}{2}$

26) $W = \frac{1}{2}(p^0 + 2p^0) \times (2V^0 - V^0) = \frac{3}{2}p^0V^0$ $\Delta U = \frac{3}{2}nR\Delta T = \frac{3}{2}[4p^0V^0 - p^0V^0] = \frac{9}{2}p^0V^0$

$\Delta Q = 6p_0V_0$

$c = \frac{\Delta Q}{n\Delta T} = \frac{6p_0V_0}{\frac{3p_0V_0}{R}} = 2R$

OR

$p \propto V$

$pV^{-1} = \text{constant}$

Now, $C = C_V + \frac{R}{1-x}$

$= \frac{3R}{2} + \frac{R}{1-(-1)} = 2R$

27) $\int dw = \int Pdv = a\int_{v_1}^{v_2} v^2dv$

$w = a\left[\frac{v^3}{3}\right]_{v_1}^{v_2} = \frac{a}{3}\left[v_2^3 - v_1^3\right]$

28)

Work done = Area under P-V graph.

29)

T = $2\pi\sqrt{\frac{\ell}{g}}$

$\ell \rightarrow$ effective length between point of suspension and COM of body

$\rightarrow$ when girl stands up , $\ell\downarrow$, so, T$\downarrow$

30) From equation : k = 5 .

As at t = 0.5 second, the particle reaches the origin for the first time.

Substituting t = 0.5 in equations provided in options.

x = 3 + 3 cos (π . $\frac{1}{2}$) = 3 → not satisfied

& x = 3 + 3 cos (2π . $\frac{1}{2}$) = 0 → satisfied.

Hence, the equation of particle is can be as given in option (4).

31) $f = \frac{v}{\lambda} = \frac{1}{\lambda}\sqrt{\frac{T}{\mu}}$

$\frac{\Delta f}{f} = \frac{1}{2} \times \frac{\Delta T}{T}$

$\Rightarrow \frac{5}{f} = \frac{1}{2} \times \frac{2}{100} = \frac{1}{100}$

f = 500 Hz

32)

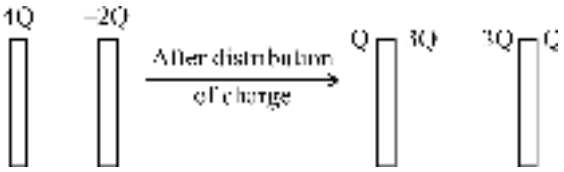
τ = P.E. sin30° = $10\sqrt{2}$

⇒ P.E = $20\sqrt{2}$ .....(1)

Now, potential energy is given by

|U| = PE cos30°

⇒ |U| = $20 \times \sqrt{2} \times \frac{\sqrt{3}}{2}$ = 24.5 J

33) 

Energy stored between two plates = $\frac{q^2}{2C}$

where, q = charge on capacitor.

 q = 3Q

So, energy store = $\frac{(3Q)^2}{2C} = \frac{9}{2}\frac{Q^2}{C}$

34) $\eta\% = \frac{R}{R+r} \times 100\%$

Case (1) :- $\frac{R}{R+r} = 0.6 \Rightarrow 0.4R = 0.6r$

$R = \frac{3}{2} r = 1.5\ r$

Case (2) :- $\eta = \frac{6R}{6R + r} = \frac{6(1.5r)}{6(1.5r) + r}$

$= \frac{9}{10} \times 100 = 90\%$

35) $\frac{R_1}{R_2} = \frac{\ell}{100 - \ell}$ .....(1)

$\frac{R_2}{R_1} = \frac{\ell + 25}{75 - \ell}$ .....(2)

On solving (1) and (2), $\ell = \frac{75}{2}$

$\therefore \frac{R_1}{R_2} = \frac{75/2}{100 - 75/2} = \frac{3}{5}$

36) $\because J = \sigma E$

$\therefore \frac{I}{A} = \frac{1}{\rho} E$

$\Rightarrow I = \frac{EA}{\rho}$

$= \frac{8.5 \times 10^{-3} \times 2 \times 10^{-6}}{1.7 \times 10^{-8}}$

$= 1A$

37)

True value of velocity $v = \frac{s}{t}$

Percentage Error in velocity $\left(\frac{\Delta v}{v} \times 100\right)$

$= \left(\frac{\Delta s}{s} \times 100\right) + \left(\frac{\Delta t}{t} \times 100\right)$

$\left(\frac{\Delta v}{v} \times 100\right) = \left(\frac{5m}{200m} \times 100\right) + \left(\frac{0.2sec}{20\,sec} \times 100\right)$

$\left(\frac{\Delta v}{v} \times 100\right)$ = (2.5%) + (1.0%) = 3.5%

38)

$\because$ 10 (VSD) = 9 (MSD)

$\therefore$ 1VSD = $\frac{9}{10}$(MSD) = $\frac{9}{10}$× 0.1 cm = 0.09 cm

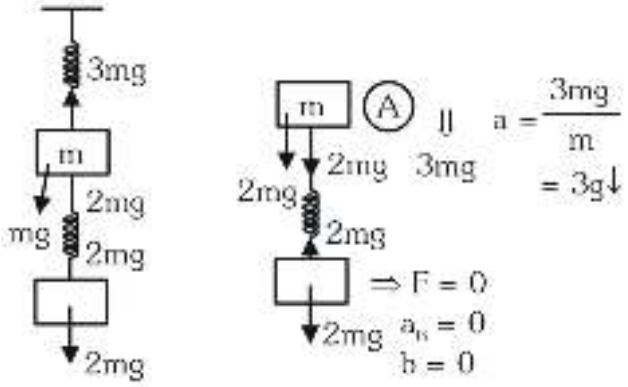
Least count L. C. = 1 MSD – 1 VSD

L. C. = 0.1 cm – 0.09 cm = 0.01 cm

Net Reading NR = MSR + VSR × LC – ZE

Net Reading NR = 1.3 cm + 2 × 0.01 cm – 0 cm

Net Reading NR = 1.3 cm + 0.02 cm – 0 cm = 1.32 cm

39) 

40)

As friction coefficient for 4 kg block is greater then that of 2 kg block, both will move together.

$$a_c = \frac{(4+2)(g\sin 30^o) - 0.3 \times 4 \times g\cos 30^o}{-0.2 \times 2 \times g\cos 30^o \over 4+2}$$

$a_c = 2.6\ m/s^2$

41)

Given $a_A = 2\alpha = 5\ m/s^2$

$\Rightarrow \alpha = 5/2\ rad/s^2$

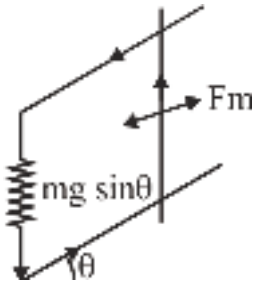
$\Rightarrow a_B = 1.(\alpha) = 5/2\ m/s^2$

42) $\frac{dQ}{dt} = \sigma eA\left(T^4 - T_0^4\right) = ms\frac{d\theta}{dt}$

so $\frac{dQ}{dt}$ will be same as 'e' is same but 's' is different so $\frac{d\theta}{dt}$ will be different.

43) q = CV
after dielectric slab
C' = KC
q remain unchanged
So V↓
V = E.d
as V↓, E↓

44) Terminal velocity is attained when magnetic force is equal to mg sin θ
 $F_m = mg\sin\theta$



or = mg sinθ

or $\left(\frac{e}{R}\right) \ell B = mg \sin\theta$

or $\frac{(B\nu_T \ell)}{R} \times \ell B = mg \sin \theta$

$\therefore \nu_T = \frac{mgR \sin\theta}{B^2\ell^2}$

45) $(\Delta x)_{medium} = (\Delta x)_{air}$

$n_1\lambda_1 = (2n_2 - 1)\frac{\lambda_2}{2}$

$10 \times \lambda_m = (2 \times 8 - 1) \times \frac{\lambda_{air}}{2}$

$10 \times \frac{\lambda_{air}}{\mu_m} = \frac{15}{2} \times \lambda_{air}$

$\mu_m = \frac{20}{15} = \frac{4}{3} = 1.33$

CHEMISTRY

46) **Explain the Question:** Identify true (T) or false (F) statements related to bonding and properties.
Concept: This question is based on polarity, bond length, hydrogen bonding, Fajan's rule, melting point trend.
Solution :
(a) XeF_4 is planar (Square Planar) but non-polar 9 dipoles cancel out)→ F
(b) Si–O < Cl–O → T (due to larger atomic radius of Si) → T
(c) Urea shows intermolecular H-bonding → F
(d) Covalent character: NaCl < CuCl < $SiCl_4$ → T
Covalent character is explained by **Fajan's Rules :** 1. Si^{4+} has the highest charge, leading to high polarization and most covalent $SiCl_4$. 2. CuCl has Cu^+ with a d^{10} (pseudo inert gas) configuration, which has higher polarizing power than Na^+ (s^2p^6 inert gas) in NaCl. Thus CuCl is more covalent than NaCl. The order is correct.
(e) Melting point decreases Hl > HBr > HCl → T (The melting point of hydrogen halides HX are primarily governed by van der walls forces, which increases which increases with molecular mass (and size).
Final answer : 3

47)

Single super pair means multiple bond.

48) NCERT Part-I (Page No. 119)

49)

$FeCl_2$. $6NH_3$ can show geometrical isomerism.

50)

Question Explanation:
The question is asking about arranging the complexes in decreasing numbers of unpaired electrons.

Concept:
Unpaired electrons depend on d-count, ligand field.

Solution:
Let's find unpaired electrons in each complex:
I. $[Fe(H_2O)_6]^{2+}$ $Fe^{2+} \rightarrow d^6$, H_2O weak field → high spin → 4unpaired electrons
II. $[Cr(NH_3)_6]^{3+}$ $Cr^{3+} \rightarrow d^3$, NH_3 strong field → 3 unpaired electrons
III. $[Ni(NH_3)_6]^{2+}$ $Ni^{2+} \rightarrow d^8$, NH_3 strong field → 2 unpaired electrons
IV. $[FeF_6]^{3-}$ $Fe^{3+} \rightarrow d^5$, F^- weak field → high spin → 5 unpaired electrons
Now arrange in decreasing (highest → lowest):
IV (5) > I (4) > II (3) > III (2)

Final Answer: Option(1)

51) NCERT-12th Part-1, Chapter - Coordination compound

52)

Explain the Question : Identify which option shows the correct decreasing order of oxidising strength.
Concept : This question is based on general properties of d-block (oxidising strength)
Solution :

A. $MnO_4^- < TcO_4^- < ReO_4^-$: Incorrect. Oxidizing power decreases down Group 7; correct order: $MnO_4^- > TcO_4^- > ReO_4^-$.

B. $CrO_4^{2-} > MoO_4^{2-} > WO_4^{2-}$: Correct. Oxidizing power decreases down Group 6; CrO_4^{2-} is strongest oxidizing agent.

C. $F_2 < Cl_2 < Br_2 < I_2$: Incorrect. Oxidizing power decreases down halogen group; correct: $F_2 > Cl_2 > Br_2 > I_2$.

D. $PbO > PbO_2$: Incorrect. PbO_2 (Pb^{4+}) is stronger oxidizing agent due to inert pair effect.

Final answer : Option (2)

53) $Na_2SO_3 + HCl \rightarrow NaCl + SO_2 + H_2O$
$Na_2S + HCl \rightarrow NaCl + H_2S$
$K_2Cr_2O_7 + SO_2 \rightarrow Cr^{+3} + SO_3$

54) NCERT-XII, Pg. # 339 (Old)

55)

Question Explanation :

We need the lanthanide ion with the maximum number of unpaired 4f electrons among Ce^{3+} to Lu^{3+}.

Concept :

Unpaired electrons depend on 4f subshell configuration ($Ce^{3+} \rightarrow Lu^{3+}$).

Solution :

$Ce^{3+} = [Xe]4f^1 \rightarrow$ 1 unpaired

$Eu^{3+} = [Xe]4f^6 \rightarrow$ 6 unpaired

$Gd^{3+} = [Xe]4f^7 \rightarrow$ 7 unpaired (half-filled, most stable)

$Lu^{3+} = [Xe]4f^{14} \rightarrow$ 0 unpaired

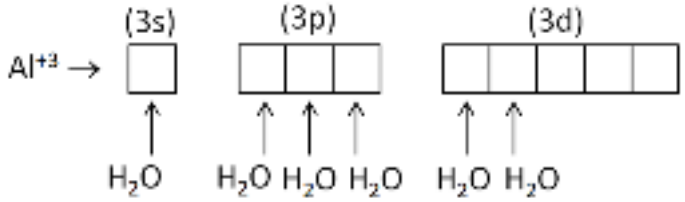
Hence, Gd^{3+} has the maximum number of unpaired electrons.

Final Answer : Option (4)

56) **Explain Question:**
Asking about incorrect statement

Concept:
Hybridisation

Solution:
$[AI(H_2O)_6]^{+3}$
$Al \rightarrow [Ne]\ 3s^2\ 3p'$
$Al^{+3} \rightarrow [Ne]\ 3s^o\ 3p^o$



Hence, hybridisation is sp^3d^2

Final Answer: Option (2)

57)

NCERT 12th, p-block

58) $_{27}M = Co^{+2} \rightarrow 3d^7$ 3 unpaired e^-.
$\mu = \sqrt{3 \times 5} = \sqrt{15}$

59)

Question Explanation: Check correctness of assertion and reason about IE trend.

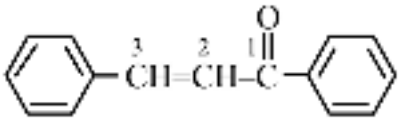
Concept: Ionisation potential in transition series.

Solution: Across transition series, effective nuclear charge increases while d-electrons shield poorly, so ionization enthalpy generally rises with small irregularities. Thus both statements are correct and Reason explains Assertion.

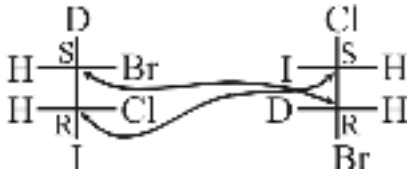
Final Answer: Option (3)

60)

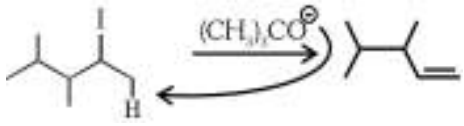
A - Cl^- gives yellow ppt - false
Reason - it gives white ppt of AgCl
D - Cl^- gives white ppt which is insoluble in NH_4OH - false
Reason - white ppt of AgCl is soluble in NH_4OH.
due to formation of soluble complex.

61) 

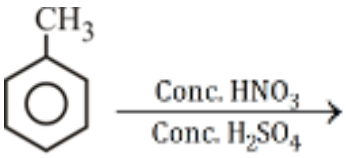
1, 3-diphenylprop-2-en-1-one

62) 

Enantiomers

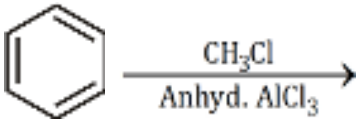
63) 

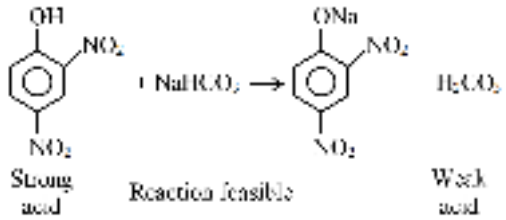
64)

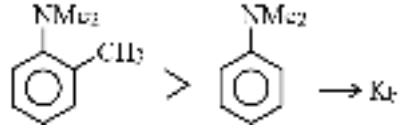
(a)  (q) Nitration

(b) CH_3–COOK $\xrightarrow{\text{electrolysis}}$ (p) Kolbe's electrolysis

(c) CH_3–Br $\xrightarrow[\text{Dryether}]{\text{Na}}$ (r) Wurtz reaction

(d)  (s) Friedal-craft alkylation

65) 

66) 

67) % of S = $\frac{32}{233} \times \frac{0.466}{0.192} \times 100$ = 33.33%

68)

Chlorine which is present at para position w.r.t –NO_2 will be replaced due to formation of stable. Carbanion intermediate.

69)

	List-I (Reaction)		**List-II (Product)**
(a)	Rosenmund reaction	(iv)	R–CHO (Aldehyde)
(b)	HVZ reaction	(iii)	R–CH(X)–COOH (a-Halo acid)
(c)	Reimer Tiemann	(i)	Salicylaldehyde (o-Hydroxybenzaldehyde)
(d)	Finkelstein reaction	(ii)	R-I (Alkyl Iodide)

70)

All a, b, c & d are correct

71) NCERT XII, Pg. # 362, 371, 372, 383
NCERT XII, Pg. # 363

72) $\text{Esterification} \propto \frac{1}{\text{Steric Hindrance}}$

73)

Question Explanation: Identify the correct statement.

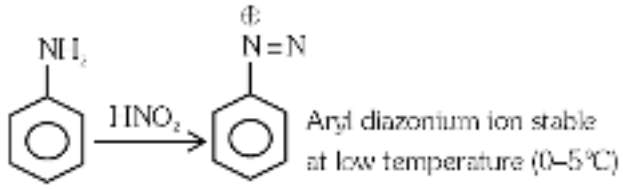
Concept: S_N1 reaction mechanism

Solution:

(A) The step-1 of S_N1 is reaction is slowest and reversible is correct since it is rate determining step where C – X bond breaks to form a carbocation intermediate and is indeed reversible.
(B) The energy for C – X bond breaking is obtained through the solvation of the halide ion with the proton of a protic solvent. Solvation by a polar protic solvent stabilizes the transition state and the resulting ions, mainly halide ion which helps to drive the reaction forward and facilitate the ionization process.
Final Answer: Option (1)

74) $R-NH_2 \xrightarrow{HNO_2} R-N_2^{\oplus}$

Alkyl diazonium ion
(unstable)



75) **Solution/Explanation :**
Glucose is aldohexose sugar which exist in pyranose form i.e. six membered ring and not in furanose form.

76)

pH = 5 ⇒ $[H^+] = 10^{-5}$
$C\alpha = 10^{-5} \Rightarrow \alpha = 10^{-5}$
$G = 4 \times 10^{-5}$ S
$G^* = \frac{\ell}{A} = \frac{15}{1} \times 100$ m^{-1}
K = G.G*

$= 4 \times 10^{-5} \times 15 \times 100$
$= 6 \times 10^{-2}$ S-m^{-1}

$\Lambda_m = \frac{K}{1000 \times M} \Rightarrow \Lambda_m = \frac{6 \times 10^{-2}}{1000 \times 1}$

$\Lambda_m = 6 \times 10^{-5}$ S-m^2/mol $\quad \alpha = \frac{\Lambda_m}{\Lambda_m^o}$

$10^{-5} = \frac{6 \times 10^{-5}}{\Lambda_m^o} \Rightarrow \Lambda_m^o = 6$ S-m^2/mol

77)

NCERT Pg # 61

78) $C_7H_{16}(\ell) + 11O_2(g) \rightarrow 7CO_2(g) + 8H_2O(\ell)$
$\Delta_{ng} = 7 - 11 = -4$

79)

$\ln k_1 = \ln A - \frac{(Ea + 20)}{300R}$ ...(1)

$\ln k_2 = \ln A - \frac{Ea}{300R}$ ...(2)

(2) – (1)

$\ln \frac{k_2}{k_1} = \frac{(Ea + 20 - Ea) \times 1000}{300R}$

$= \frac{20 \times 1000}{300 \times 8.3}$
$= 8$

80) $(Eq.)_{K_2Cr_2O_7} = (Eq.)_{\text{Mohr's salt}}$
$200 \times x \times 6 = 750 \times 0.6 \times 1$
$x = 0.375$

81) $P_{N_2} = K_H . X_{N_2}$

$10 \times 0.8 = \left(\frac{6.5 \times 10^7}{760}\right) X_{N_2}$

$X_{N_2} = 9.35 \times 10^{-5}$

82)

NCERT Pg # 74

83) Let moles of $(NH_4)_2SO_4$ = x
then

$$pOH = pKb + \log \frac{[NH_4^+]}{[NH_3]}$$

$$5.74 = 4.74 + \log \left(\frac{2x}{0.2}\right)$$

$$1 = \log \frac{2x}{0.2} \Rightarrow 10 = \frac{2x}{0.2}$$

x = 1

then mass of $(NH_4)_2SO_4$ = 1 × 132 g = 132 g

84) $H_2(g) + 2Fe^{+3}(aq) \xrightarrow{n=2} 2H^+(aq) + 2Fe^{+2}(aq)$

$$E_{cell} = 0.77 - \frac{0.06}{2} \log \left(\frac{[Fe^{+2}]}{[Fe^{+3}]}\right)^2$$

$$= 0.77 - \frac{0.06}{2} \log (10)^2$$

= 0.77 – 0.06

= 0.71 V

85) $\Delta H = nC_p.m\ \Delta T$

= 5 (20.3) (415–335) = 8.12 KJ

86) Old NCERT Pg. # 212

87)

NCERT Pg # 188

88) $\because \Delta T_f \propto i$

and if ΔT_f is lowest, then T_f will be highest.

89)

x + O_2 → CO_2 + H_2O

0.01 mol 0.9 g

$$n_H = \frac{0.9}{18} \times 2$$

mass % of H = 10

$$\frac{\left(\frac{0.9 \times 2}{18}\right)}{0.01 \times M_w} \times 100 = 10$$

$$M_w = \frac{0.9 \times 2 \times 100}{0.01 \times 18 \times 10}$$

= 100 g mol^{-1}

90)

$HCl(g) + 25aq. \rightarrow HCl.25aq.$ ...(1)

$\Delta H_1 = -72.03 \text{ KJ mol}^{-1}$

$HCl(g) + 40aq. \rightarrow HCl.40aq.$ ...(2)

$\Delta H_2 = -72.79 \text{ KJ mol}^{-1}$

(2) – (1)

$HCl.25aq. + 15aq. \rightarrow HCl.40aq.$

$\Delta H = -72.79 + 72.03$

$= -0.76 \text{ KJ mol}^{-1}$

BIOLOGY

91) NCERT Pg.#69-70

92) NCERT, Pg # 67

93) NCERT Pg.#60,61,69

94) NCERT Page No. - 76

95) NCERT, Pg # 62

96) NCERT, Pg. # 101

97) NCERT Pg. # 92 (E), 101,102 (H)

98) NCERT (XII)th Pg. # 80, 81

99) NCERT Pg. # 88, 89

100) NCERT Pg. # 96

101) NCERT Pg. # 104

102) NCERT Pg. # 138

103) NCERT Pg. # 149

104) Product and sites of the components of aerobic breakdown

105) NCERT, Pg. # 159, 160, 161

106) NCERT XI Pg No. # 230/231/232

107) NCERT Pg. # 175-178

108)

NCERT-XI_Pg. # 170

109) NCERT Pg# No. 114

110) NCERT, Pg. # 117

111) NCERT Pg.#106,108,109,110

112) NCERT Pg # 108

113)

NCERT (XII) Page No. 11, 12

114) NCERT, 18, 19, 20

115) NCERT XII Pg. No. # 1

116) NCERT page No. 73,74

117) NCERT Page No. - 75

118) NCERT Pg # 60, 61

119) NCERT Page No. 64

120) NCERT_Pg. No. # 63

121) NCERT Pg.#32,33

122) NCERT Pg. # 30

123) NCERT Pg. no. 24, 26

124) NCERT Page No. 18

125) NCERT Page #14,16,18,21

126)

NCERT Pg. # 11

127) Module - 1

128) NCERT, Pg.# 198, 199, 200

129) NCERT-XII, Pg. No. # 201, 202

130) NCERT Page #195, 196(E)
NCERT Page #213, 214(H)

131) NCERT XII Pg. # 212, 213

132) NCERT Pg # 206

133) NCERT, XII Pg. No. # 209

134) NCERT, XII Pg. No. # 223

135)

NCERT, XII Pg. No. # 221

136) NCERT Pg. # 44

137) Ornithorhynchus is oviparous, not viviparous

138) NCERT, Pg. # 49, 50

139) NCERT, Pg. # 47

140) NCERT Pg. # 101

141) NCERT Pg.# 12

142) NCERT XI (NEW) PAGE NO. 80,82,83,84

143) NCERT Pg. # 80

144) NCERT Pg. # 200, 201

145)

NCERT-XI_Pg. # 278, 279

146) NCERT, Pg. # 186-187

147) NCERT XI, Pg. # 190

148) NCERT Pg.#208

149) NCERT-XI, Pg., # 29

150) NCERT (XII) ; Page No. # 317 (E)
NCERT (XII) ; Page No. # 317 (H)

151) NCERT (XII) ; Page No. # 317 (E)
NCERT (XII) ; Page No. # 316 (H)

152) NCERT Pg. # 244

153) NCERT, Pg. # 242

154) NCERT Pg # 248

155)

NCERT Pg. # 242

156)

NCERT Pg. # 225

157) NCERT-Pg. # 21

158)

NCERT Pg. # 64
Fallopian tube

159) NCERT, Pg. # 37

160)

NCERT Pg.# 30, 31, 37

161) NCERT Pg.# 27

162) NCERT, Pg. # 31,32

163) NCERT Pg. # 61

164) NCERT Pg. # 137

165) NCERT, Pg. # 140

166) NCERT, Pg. # 131-133

167) NCERT Pg. # 134-135

168) NCERT, Pg. # 116

169) NCERT Pg. # 140, 141

170) The Correct answer is (3).

171) NCERT-XI Pg # 98

172) NCERT Pg # 96,97

173) New NCERT Pg. # 126, 127

174)

NCERT-XI, Pg. # 100

175) NCERT Pg # 202-203

176) NCERT Pg. No. # 164 [E]
NCERT Pg. No. # 181 [H]

177) NCERT, Pg. # 169

178)

NCERT-XII Pg. # 179

179) NCERT Pg # 213

180)

NCERT-XII Pg. # 151, 158